

SBD - Project

Is better to take performance metrics from different dbs then to take from different pc configs.

MariaDB -

PostgreDB -

MySQL -

SQLServer -

Oracle -

Db2 -

Also test different bd file locations

- HDD

- SSD

- External Drive

Also Test on different hardware

tpc-h

Show VARIABLES LIKE 'max_connections';

Set GLOBAL max_connections = **500**;

Max Connections

Buffer pool size

Log Buffer size

Change also the virtual users

Lauch the hammerdb:

```
open -a XQuartz
```

Then open XQuartz terminal:

```
xhost +
```

Then in VM terminal:

```
export DISPLAY=host.docker.internal:0.0
```

```
/opt/HammerDB-5.0/hammerdb
```

ip of machine: host.docker.internal

Tests:

MariaDB:

Test Case	max_connections	buffer_pool_size	log_buffer_size	virtual_users
Small	151	1G	16M	2
Medium	151	1G	16M	5
Large	200	1G	16M	2
Max CPU	200	1G	16M	5

1:

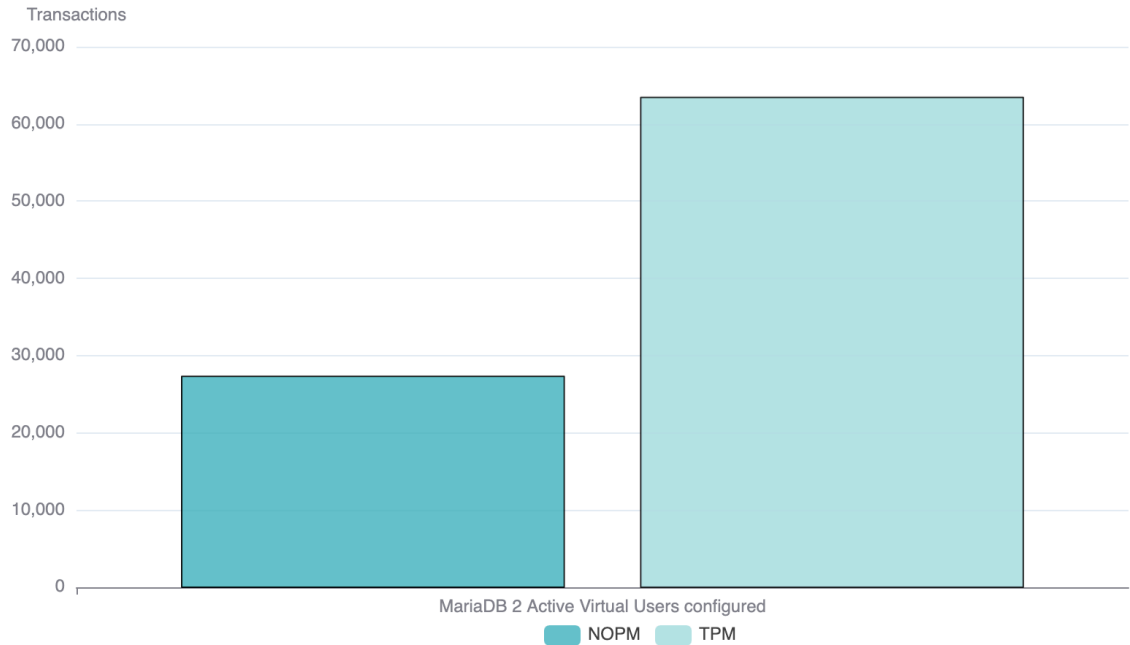
DATES:

18/05/2025
Virtual Users: 2
Wheare houses: 10
Iterations: 1
NOPM:27317
TPM:63435

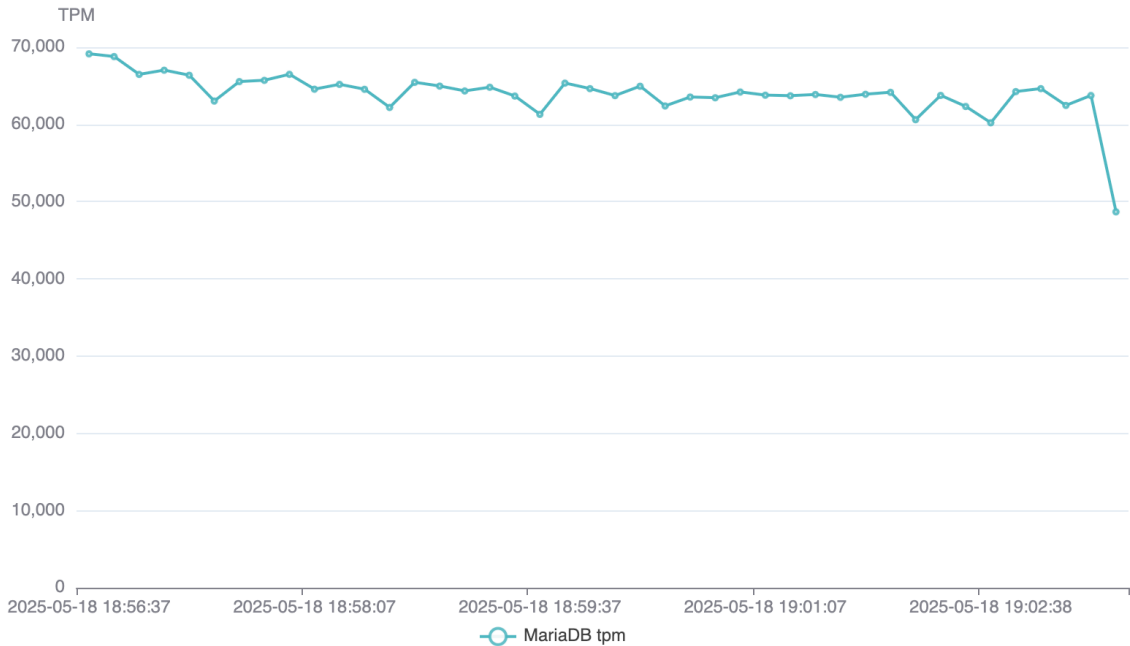
OS: UBUNTU
Drive: SSD
Max_connections = 151
Buffer Pool size = 1G
Log Buffer size = 16MB

HDB Version	Database	Benchmark	NOPM	TPM	Active VU
v5.0	MariaDB	TPROC-C	27317	63435	2

Result



Transaction Count



[Job Index](#)

2:

DATES:

18/05/2025

Virtual Users: 5

Where houses: 10

Iterations: 1

NOPM: 32809

TPM: 75921

OS: UBUNTU

Drive: SSD

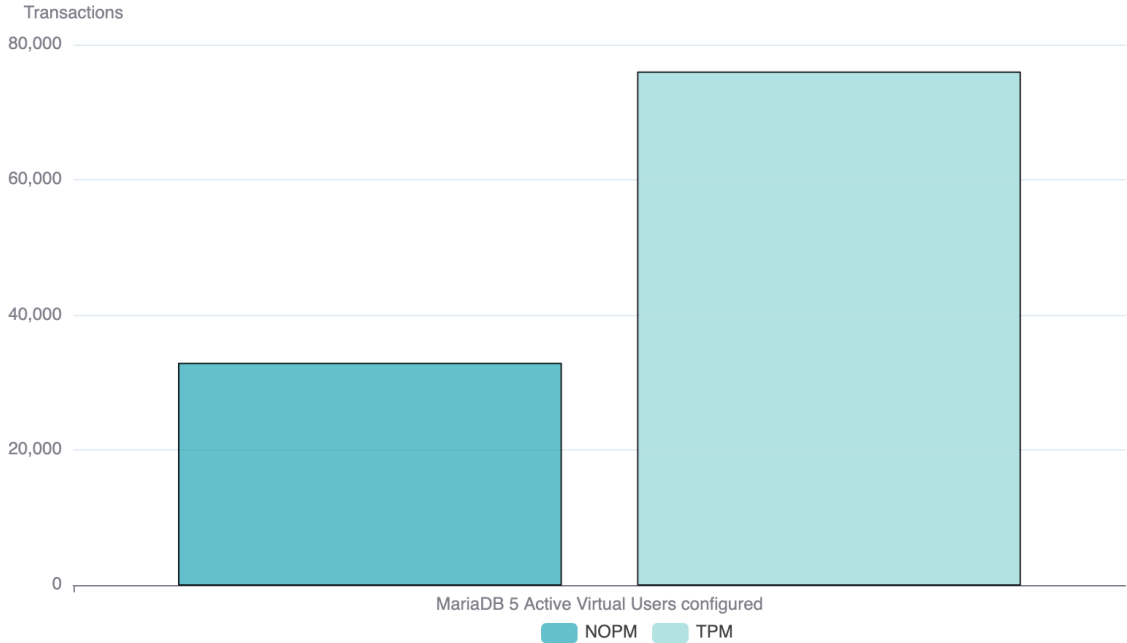
Max_connections = 151

Buffer Pool size = 1G

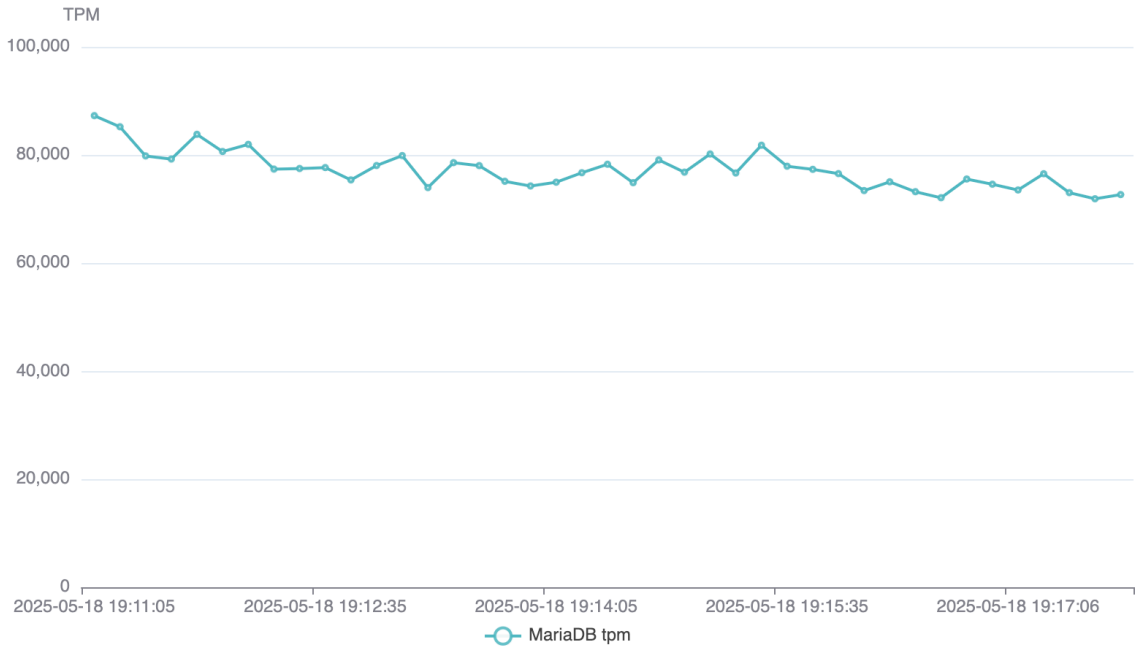
Log Buffer size = 16MB

HDB Version	Database	Benchmark	NOPM	TPM	Active VU
v5.0	MariaDB	TPROC-C	32809	75921	5

Result



Transaction Count



[Job Index](#)

3:

DATES:

19/05/2025

Virtual Users: 2

Where houses: 10

Iterations: 1

NOPM: 27250

TPM: 63390

OS: UBUNTU

Drive: SSD

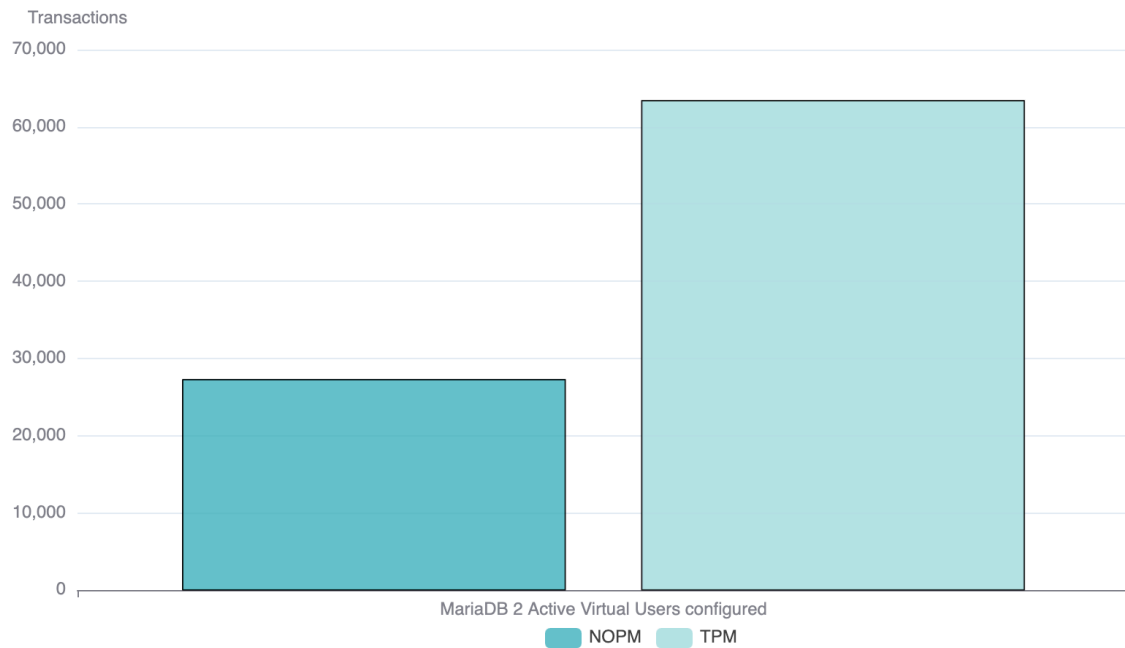
Max_connections = 200

Buffer Pool size = 1G

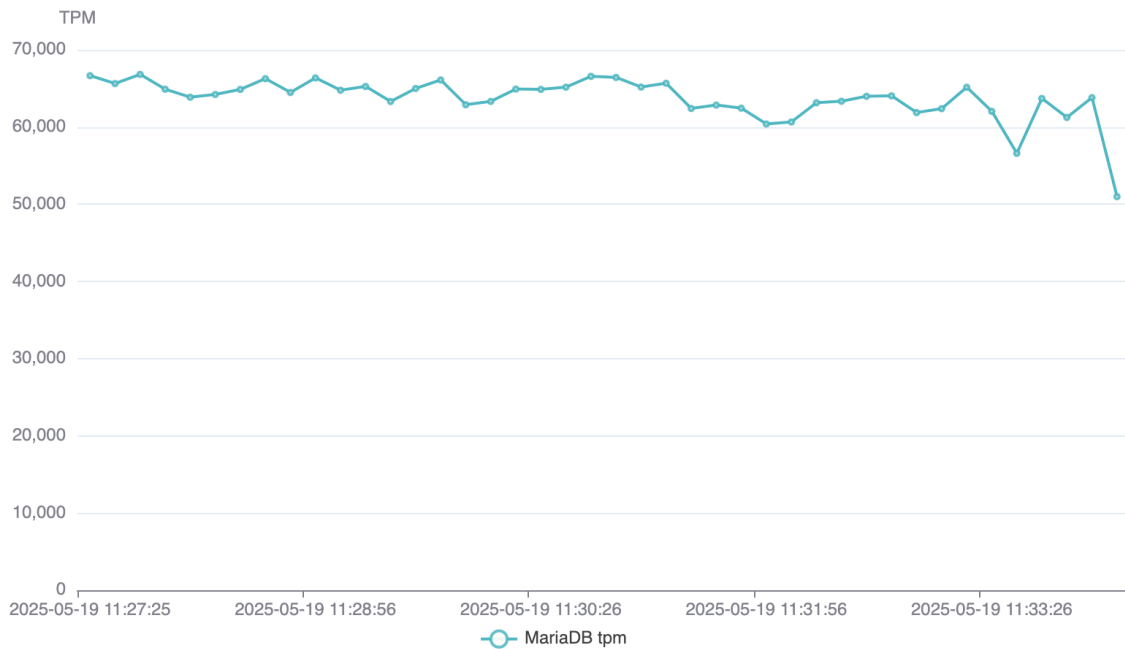
Log Buffer size = 16MB

HDB Version	Database	Benchmark	NOPM	TPM	Active VU
v5.0	MariaDB	TPROC-C	27250	63390	2

Result



Transaction Count



[Job Index](#)

4:

DATES:

19/05/2025

Virtual Users: 5

Where houses: 10

Iterations: 1

NOPM: 34419

TPM: 79599

OS: UBUNTU

Drive: SSD

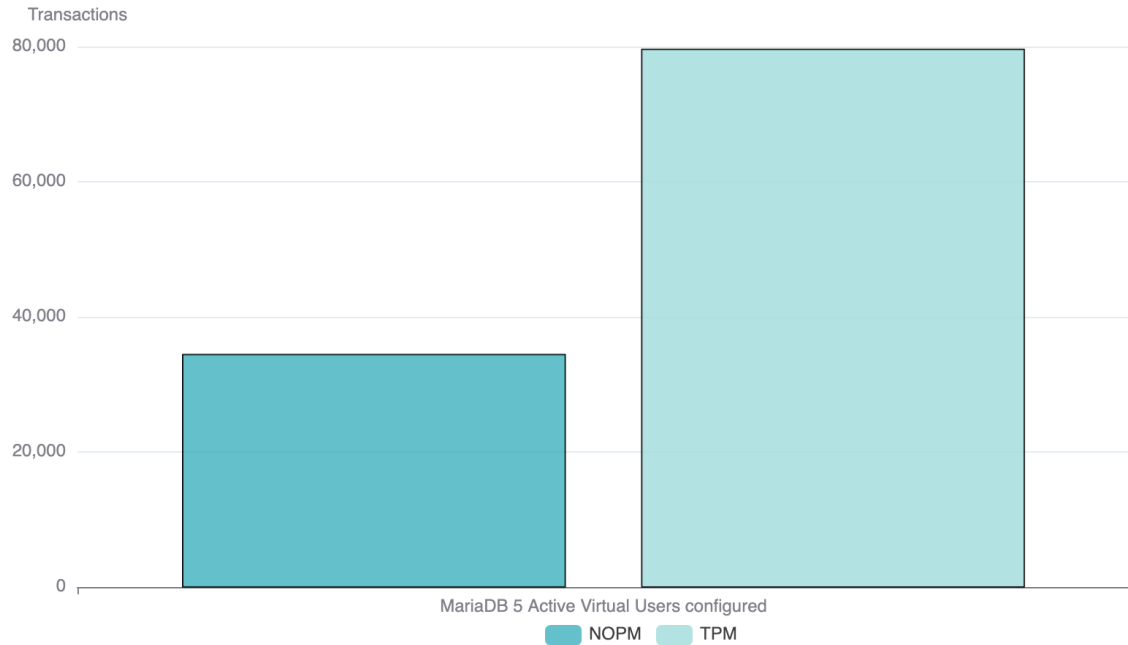
Max_connections = 200

Buffer Pool size = 1G

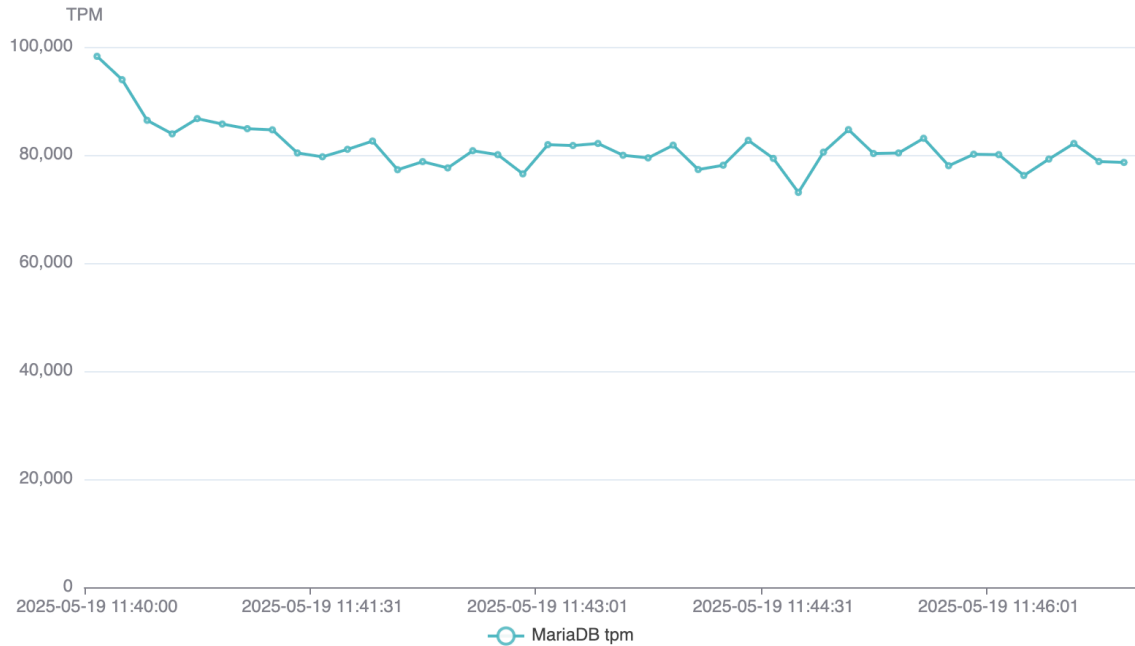
Log Buffer size = 16MB

HDB Version	Database	Benchmark	NOPM	TPM	Active VU
v5.0	MariaDB	TPROC-C	34419	79599	5

Result



Transaction Count



[Job Index](#)

Now let's compare with others:

Test Case	max_connections	buffer_pool_size	log_buffer_size	virtual_users
Small	100	256M	8M	5

Test Case	max_connections	buffer_pool_size	log_buffer_size	virtual_users
Medium	200	512M	16M	10
Large	500	1G	32M	20
Max CPU	500	2G	64M	50
I/O Test	200	256M	8M	20
Memory	200	2G	64M	10
Latency	100	128M	8M	5
Stress	1000	4G	128M	100

1:

DATES:

19/05/2025

Virtual Users: 5

Where houses: 5

Iterations: 1

NOPM: 52171

TPM: 120634

OS: UBUNTU

Drive: SSD

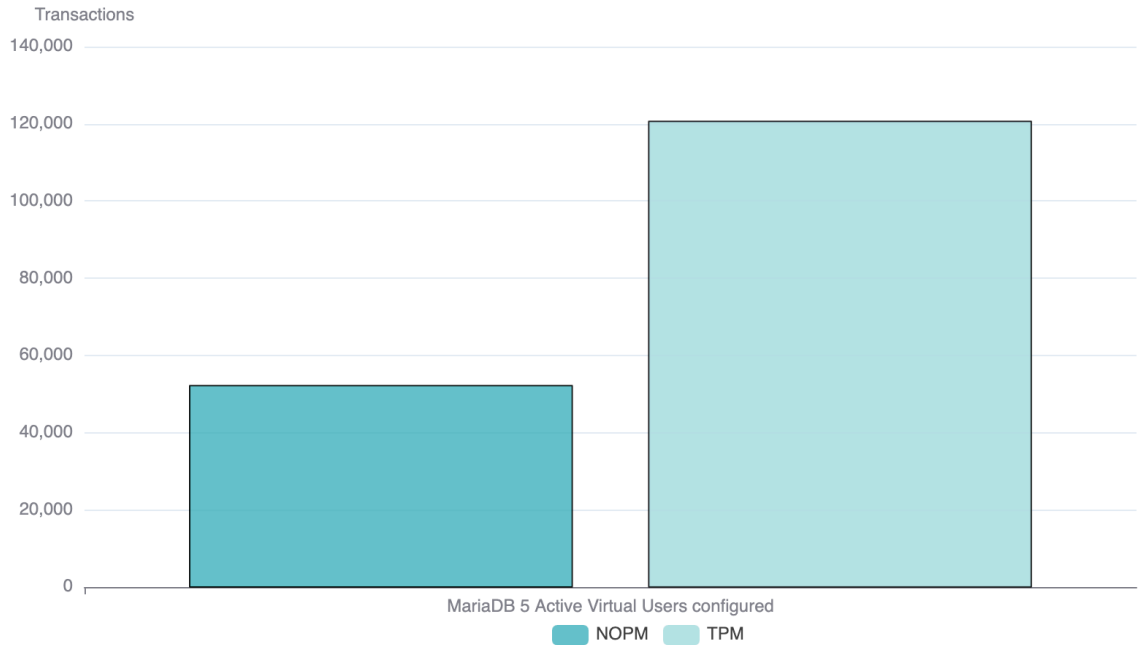
Max_connections = 100

Buffer Pool size = 256MB

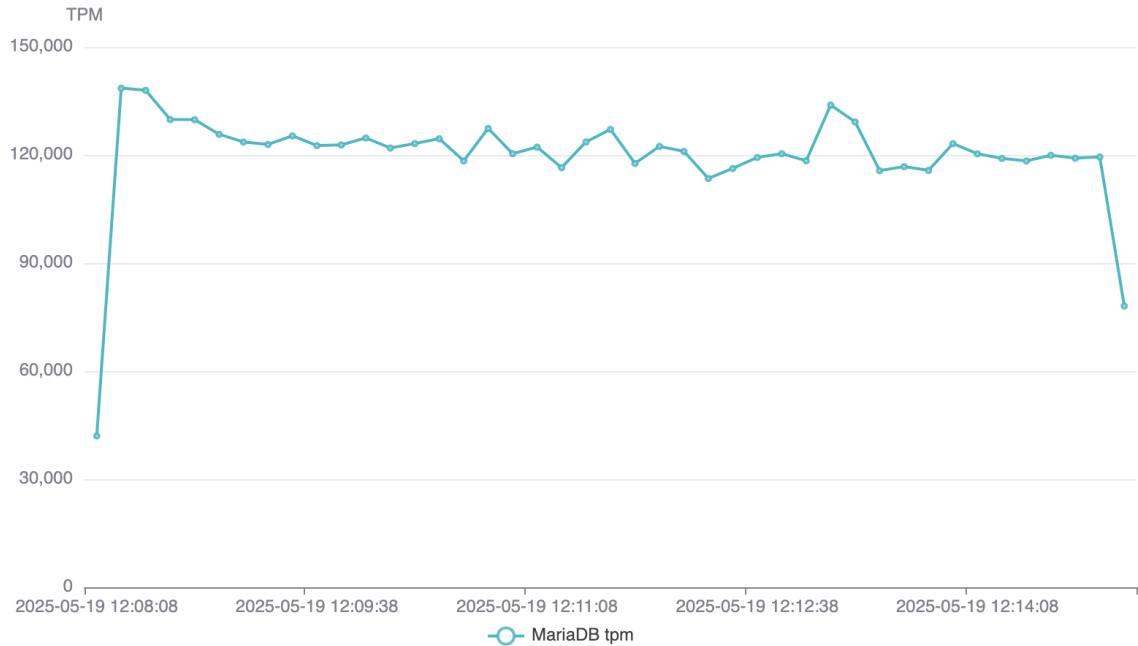
Log Buffer size = 8MB

HDB Version	Database	Benchmark	NOPM	TPM	Active VU
v5.0	MariaDB	TPROC-C	52171	120634	5

Result



Transaction Count



[Job Index](#)

2:

DATES:

19/05/2025

Virtual Users: 10

Warehouses: 10

Iterations: 1

NOPM: 81529

TPM: 187858

OS: UBUNTU

Drive: SSD

Max_connections = 200

Buffer Pool size = 512MB

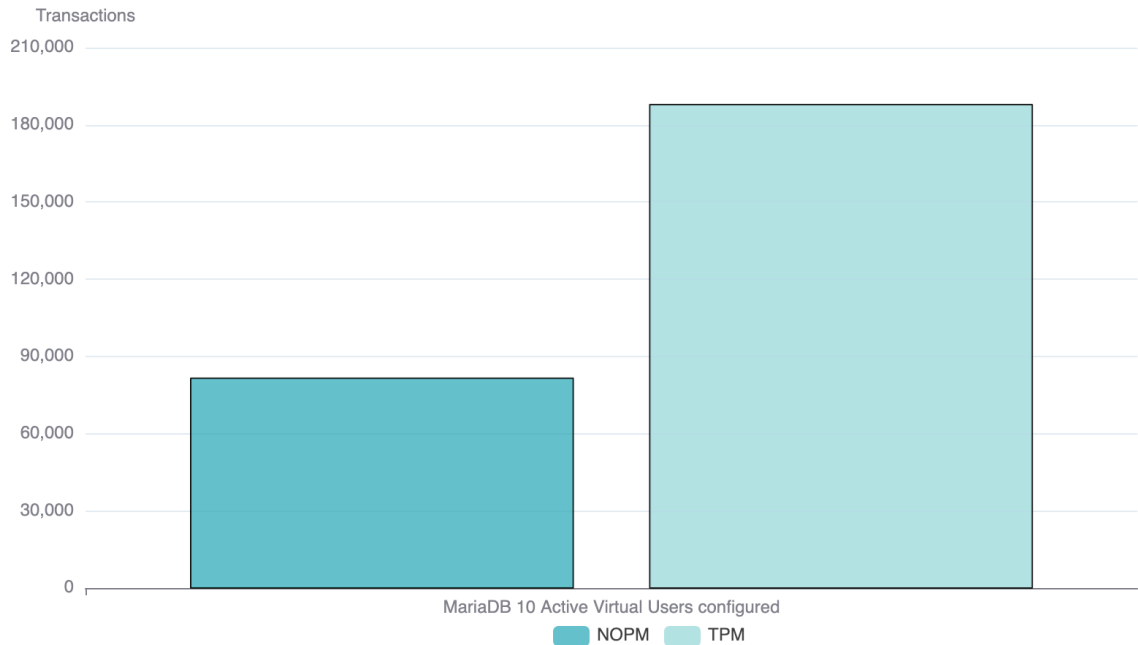
Log Buffer size = 16MB



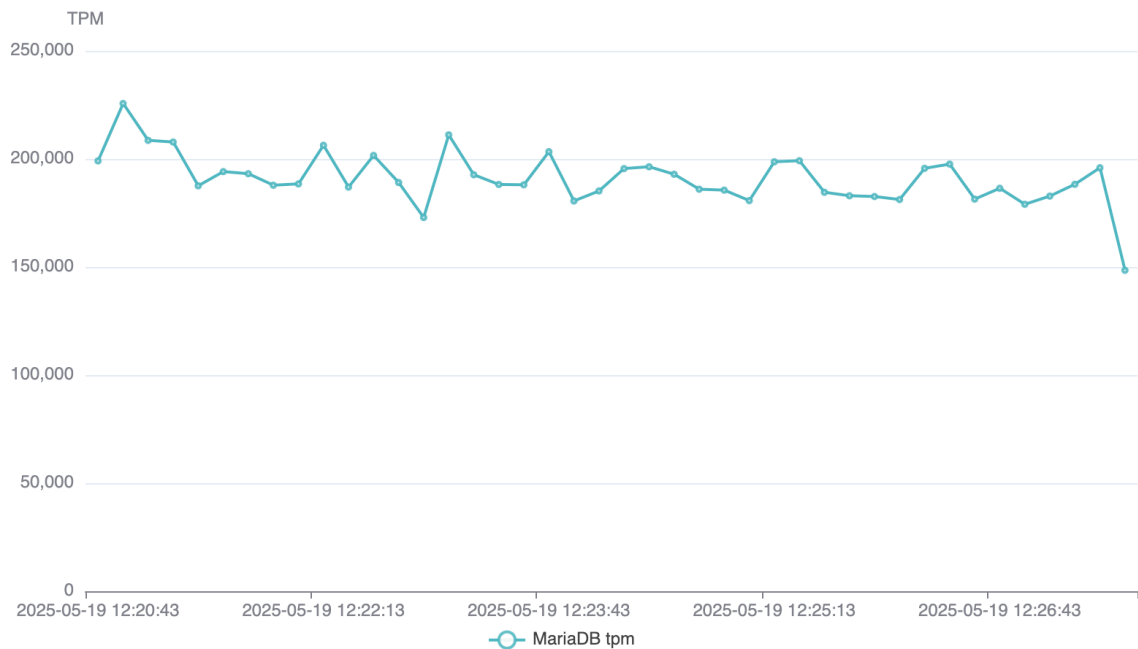
Job 682B13FF51E903E283634363 TPROC-C Summary 2025-05-19 12:20:31

HDB Version	Database	Benchmark	NOPM	TPM	Active VU
v5.0	MariaDB	TPROC-C	81529	187858	10

Result



Transaction Count



[Job Index](#)

3:

DATES:

19/05/2025

Virtual Users: 20

Warehouses: 20

Iterations: 1

NOPM: 105417

TPM: 241170

OS: UBUNTU

Drive: SSD

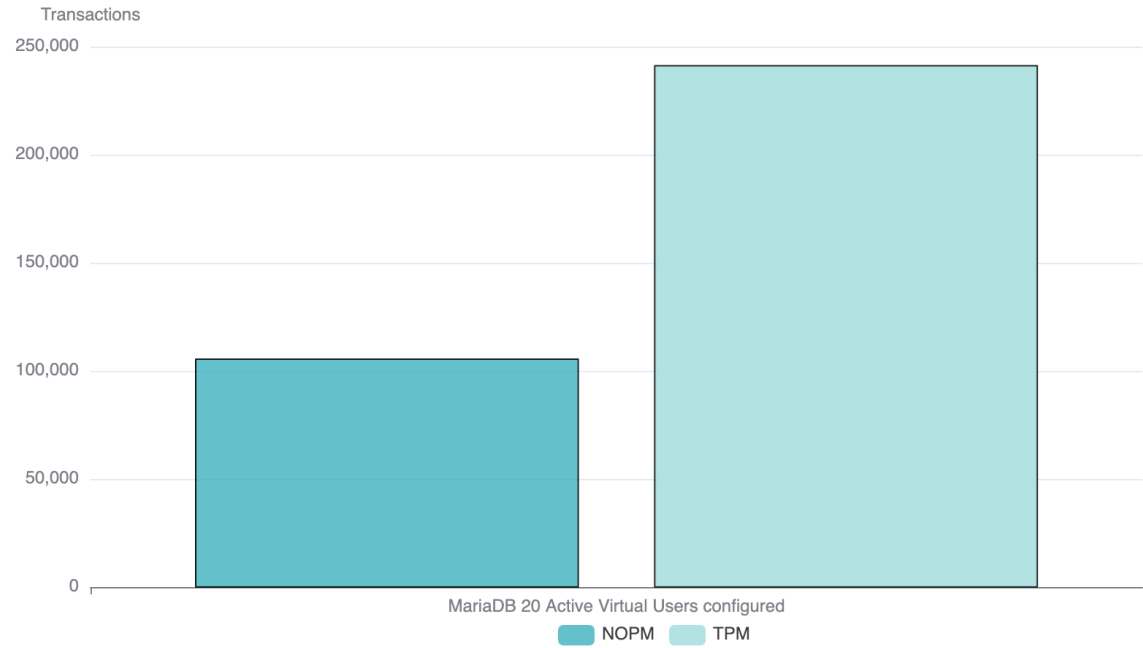
Max_connections = 500

Buffer Pool size = 1G

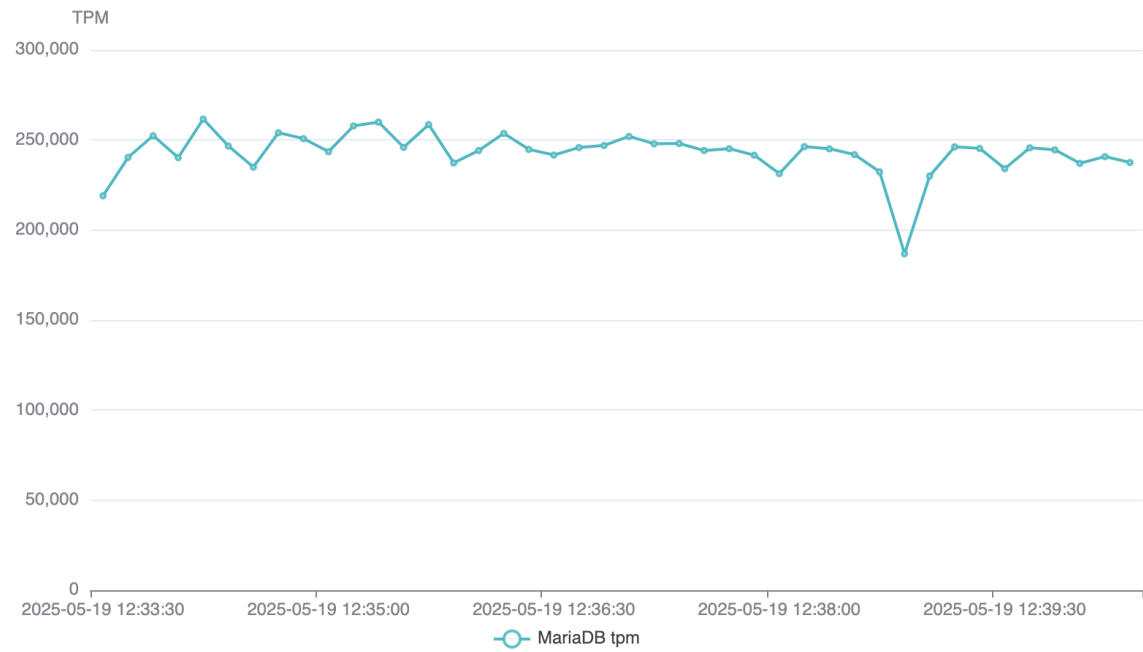
Log Buffer size = 32MB

HDB Version	Database	Benchmark	NOPM	TPM	Active VU
v5.0	MariaDB	TPROC-C	105417	241170	20

Result



Transaction Count



[Job Index](#)

4:

DATES:

19/05/2025

Virtual Users: 50

Warehouses: 20

Iterations: 1

NOPM: 44438

TPM: 103331

OS: UBUNTU

Drive: SSD

Max_connections = 500

Buffer Pool size = 2G

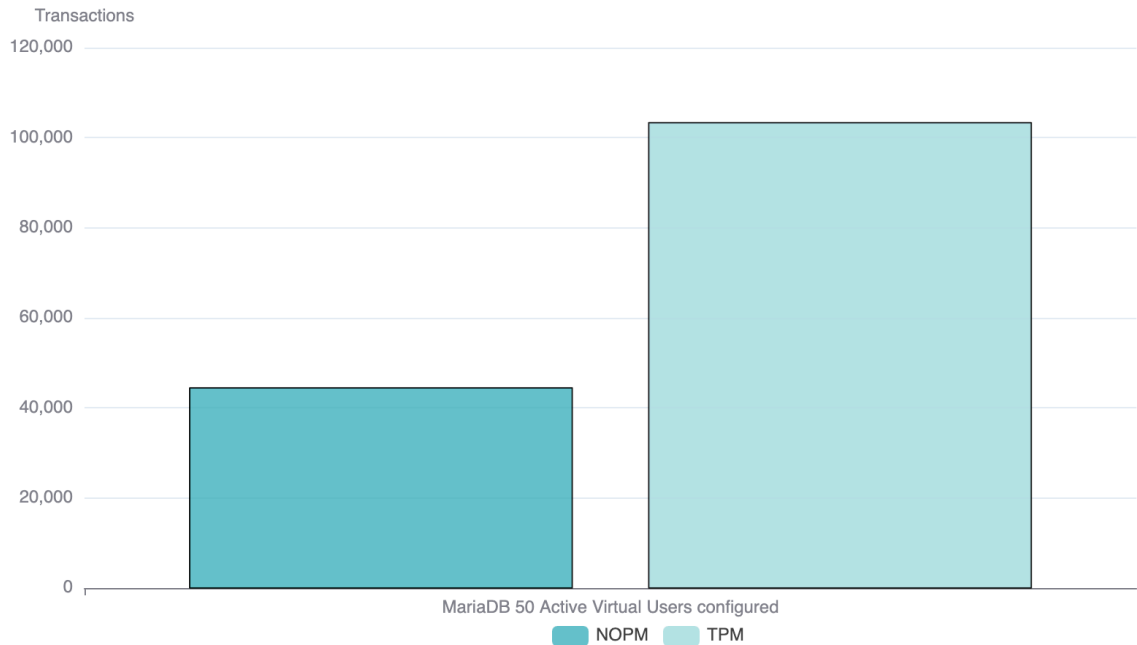
Log Buffer size = 64MB



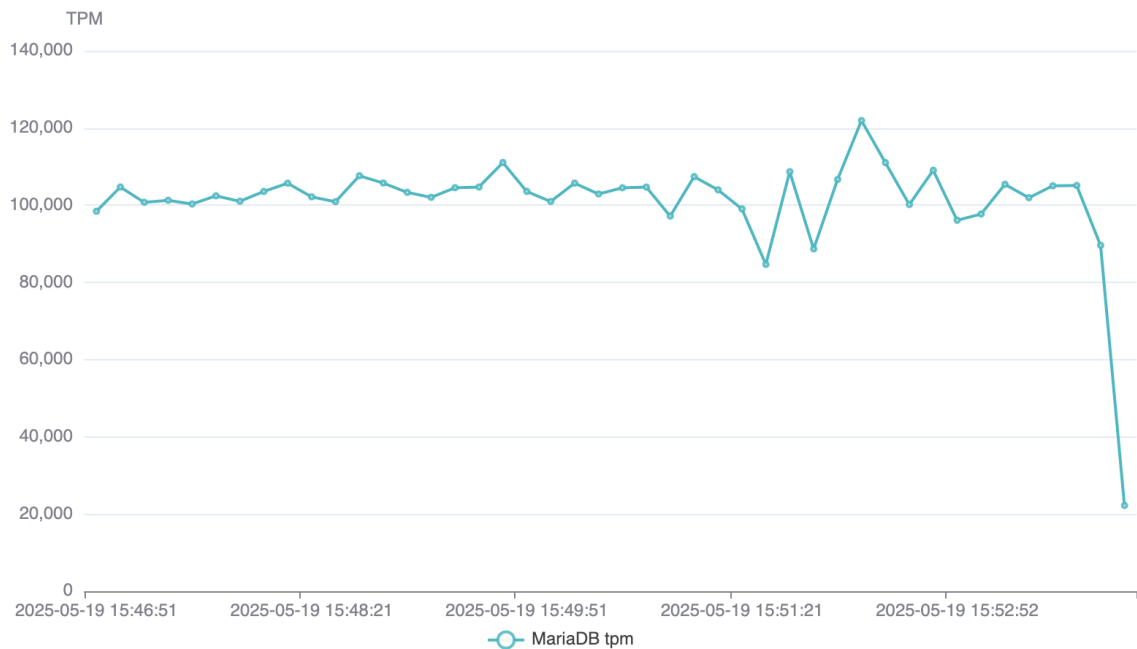
Job 682B4450331E03E253738373 TPROC-C Summary 2025-05-19 15:46:40

HDB Version	Database	Benchmark	NOPM	TPM	Active VU
v5.0	MariaDB	TPROC-C	44438	103331	50

Result



Transaction Count



[Job Index](#)

5:

DATES:

19/05/2025

Virtual Users: 20

Warehouses: 5

Iterations: 1

NOPM: 73619

TPM: 170954

OS: UBUNTU

Drive: SSD

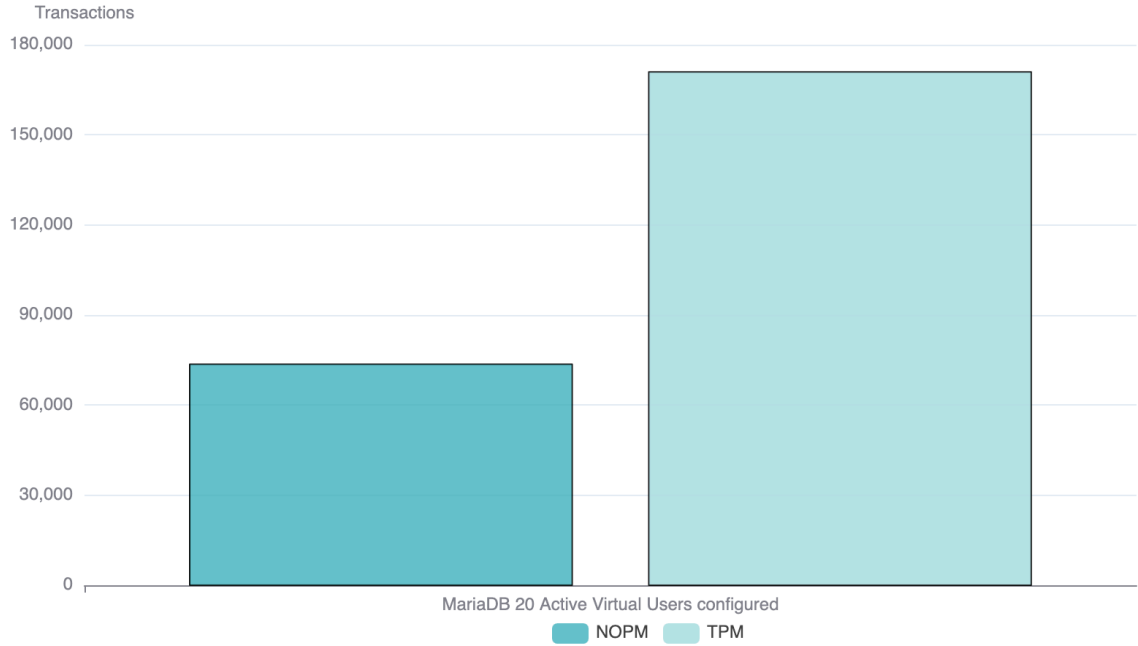
Max_connections = 200

Buffer Pool size = 256MB

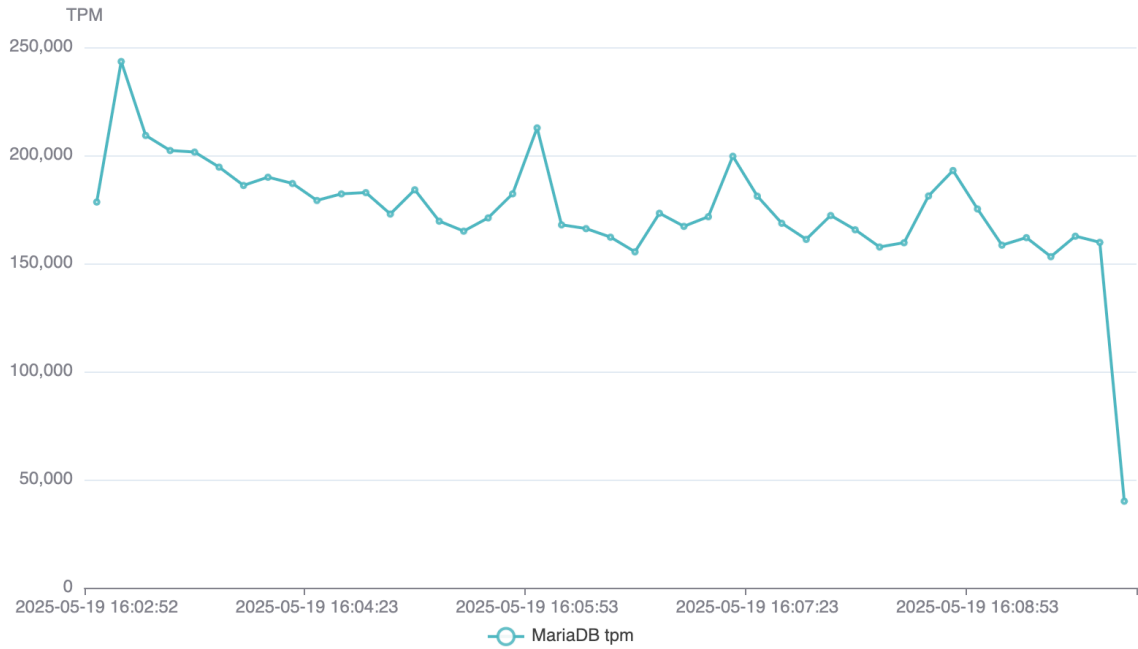
Log Buffer size = 8MB

HDB Version	Database	Benchmark	NOPM	TPM	Active VU
v5.0	MariaDB	TPROC-C	73619	170954	20

Result



Transaction Count



[Job Index](#)

6:

DATES:

19/05/2025

Virtual Users: 10

Warehouses: 10

Iterations: 1

NOPM: 42691

TPM: 98935

OS: UBUNTU

Drive: SSD

Max_connections = 200

Buffer Pool size = 2G

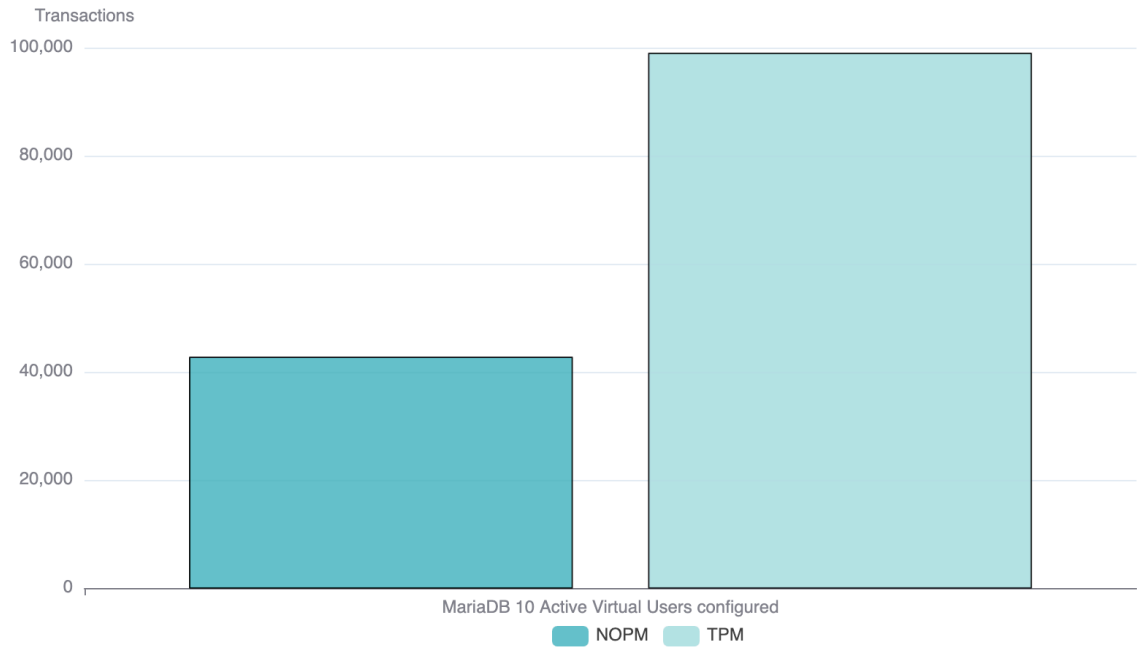
Log Buffer size = 64MB



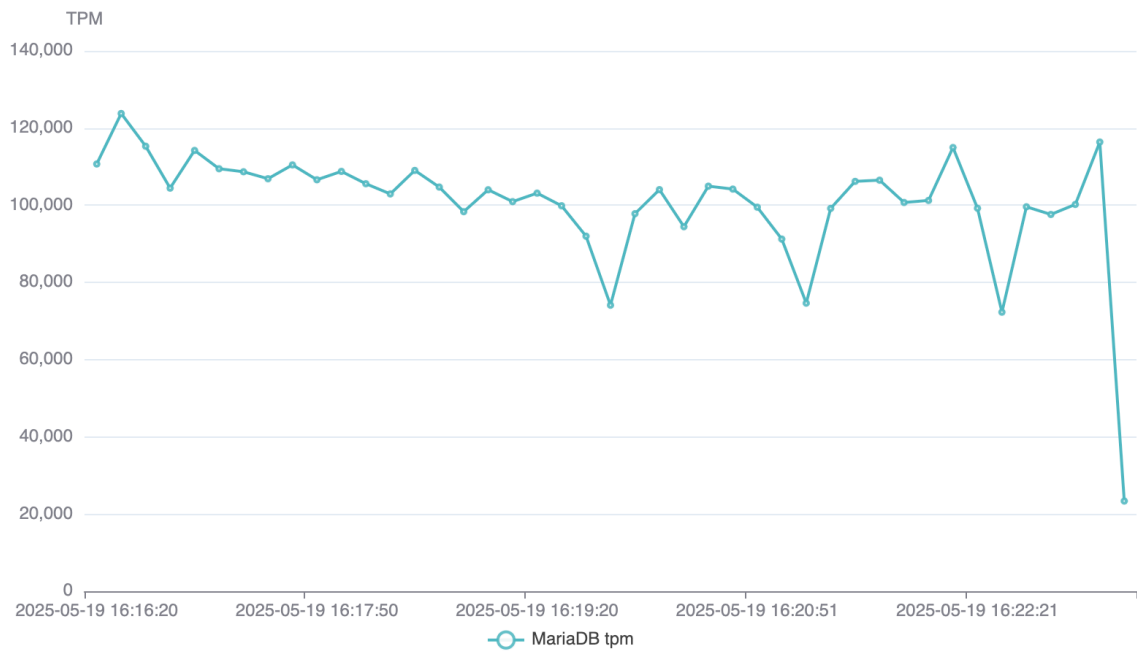
Job 682B4B399C9503E263435373 TPROC-C Summary 2025-05-19 16:16:09

HDB Version	Database	Benchmark	NOPM	TPM	Active VU
v5.0	MariaDB	TPROC-C	42691	98935	10

Result



Transaction Count



[Job Index](#)

7:

DATES:

19/05/2025

Virtual Users: 5

Warehouses: 20

Iterations: 1

NOPM: 41251

TPM: 95860

OS: UBUNTU

Drive: SSD

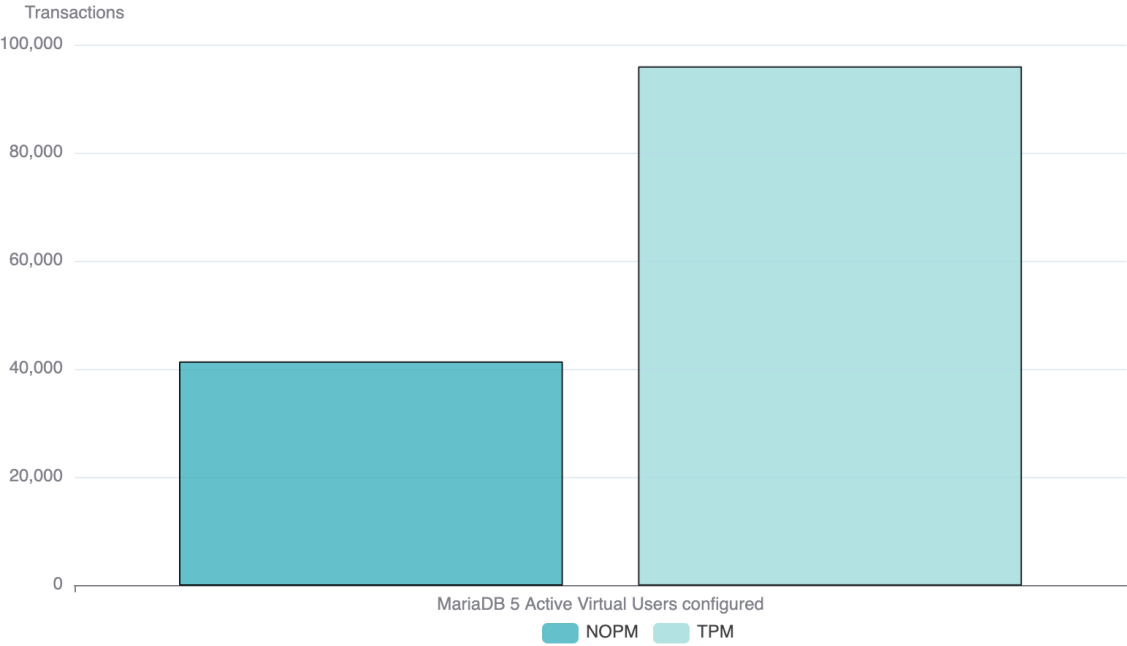
Max_connections = 100

Buffer Pool size = 128MB

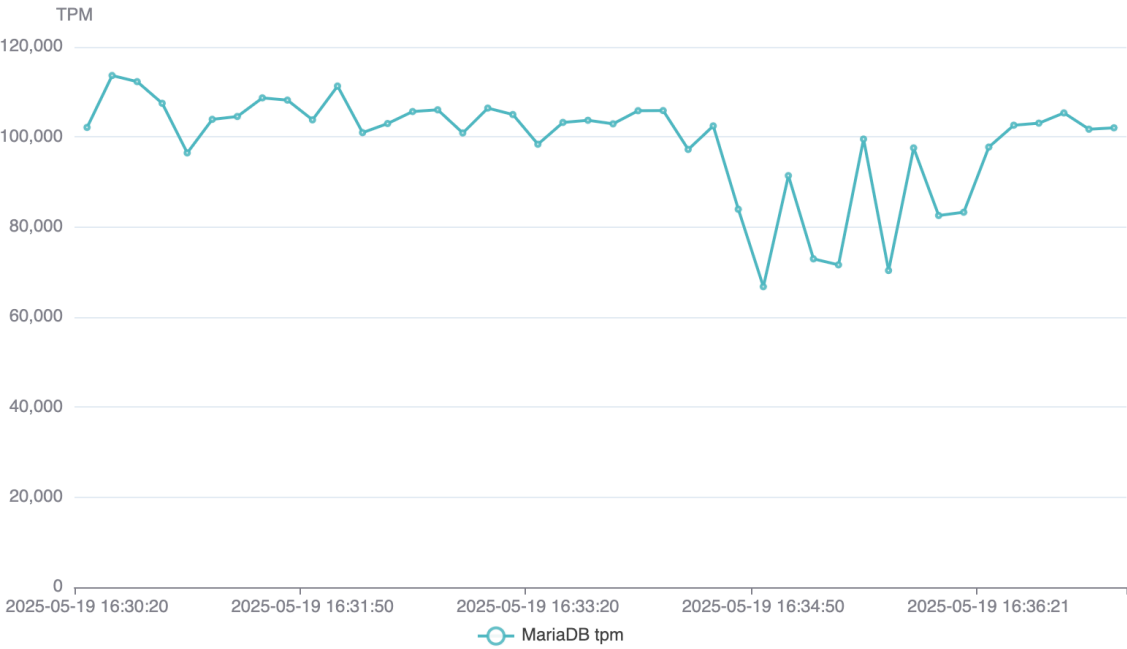
Log Buffer size = 8MB

HDB Version	Database	Benchmark	NOPM	TPM	Active VU
v5.0	MariaDB	TPROC-C	41251	95860	5

Result



Transaction Count



[Job Index](#)

8:

DATES:

19/05/2025

Virtual Users: 100

Warehouses: 50

Iterations: 1

NOPM: 28298

TPM: 65671

OS: UBUNTU

Drive: SSD

Max_connections = 1000

Buffer Pool size = 4G

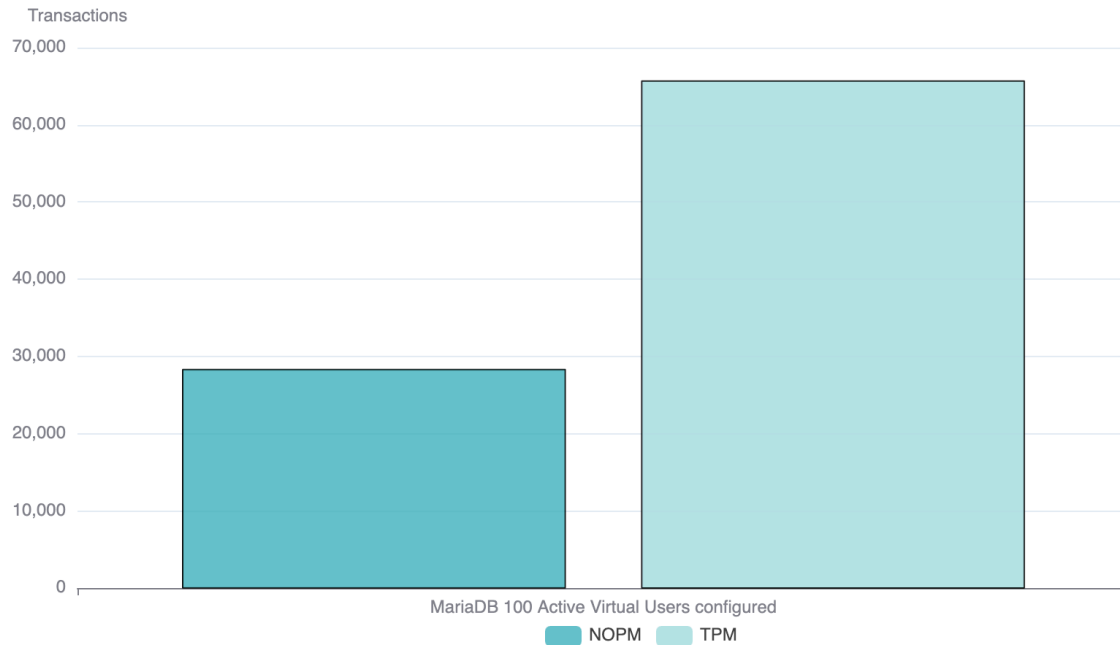
Log Buffer size = 128MB



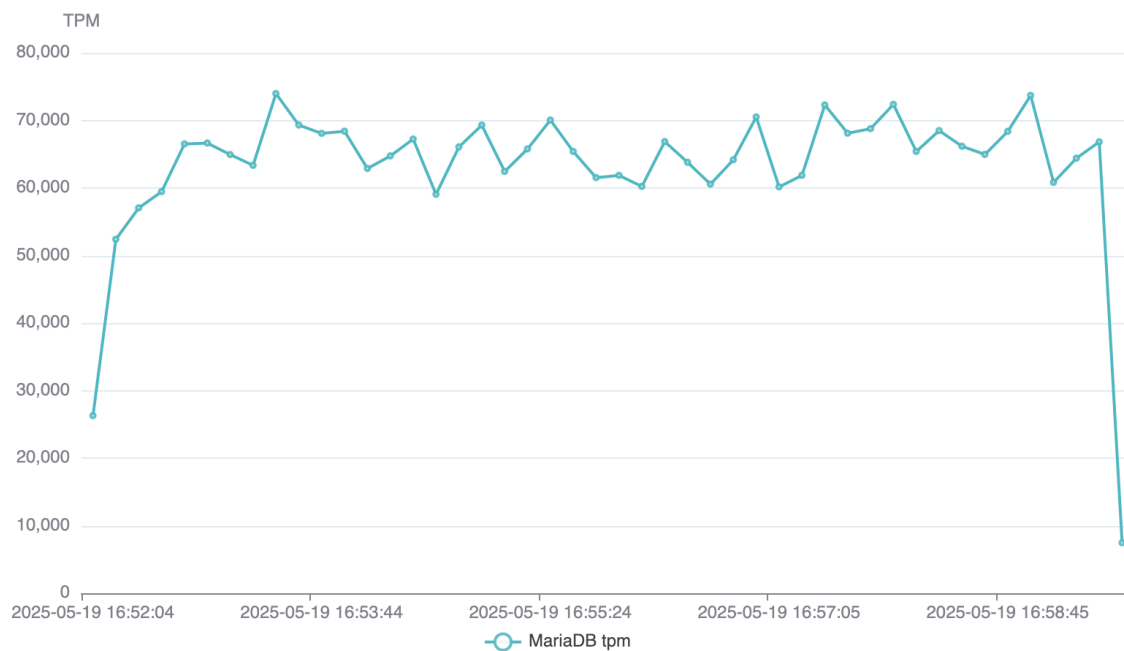
Job 682B53981C4E03E283134383 TPROC-C Summary 2025-05-19 16:51:52

HDB Version	Database	Benchmark	NOPM	TPM	Active VU
v5.0	MariaDB	TPROC-C	28298	65671	100

Result



Transaction Count



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PostgreSQL:

```
docker run -d --name postgres-bench  
-e POSTGRES_PASSWORD=root  
-e POSTGRES_DB=tpcc
```

-p 5432:5432

postgres:16

-c max_connections=151

-c shared_buffers=1GB

-c wal_buffers=16MB

CREATE DATABASE tpcc;

CREATE USER tpcc WITH PASSWORD 'root';

GRANT ALL PRIVILEGES ON DATABASE tpcc TO tpcc;

ERROR: database "tpcc" already exists

CREATE ROLE

GRANT

Test Case	max_connections	buffer_pool_size	log_buffer_size	virtual_users
Small	151	1G	16M	2
Medium	151	1G	16M	5
Large	200	1G	16M	2
Max CPU	200	1G	16M	5

1:

DATES:

18/05/2025

Virtual Users: 2

Where houses: 10

Iterations: 1

NOPM:

TPM:

OS: UBUNTU

Drive: SSD

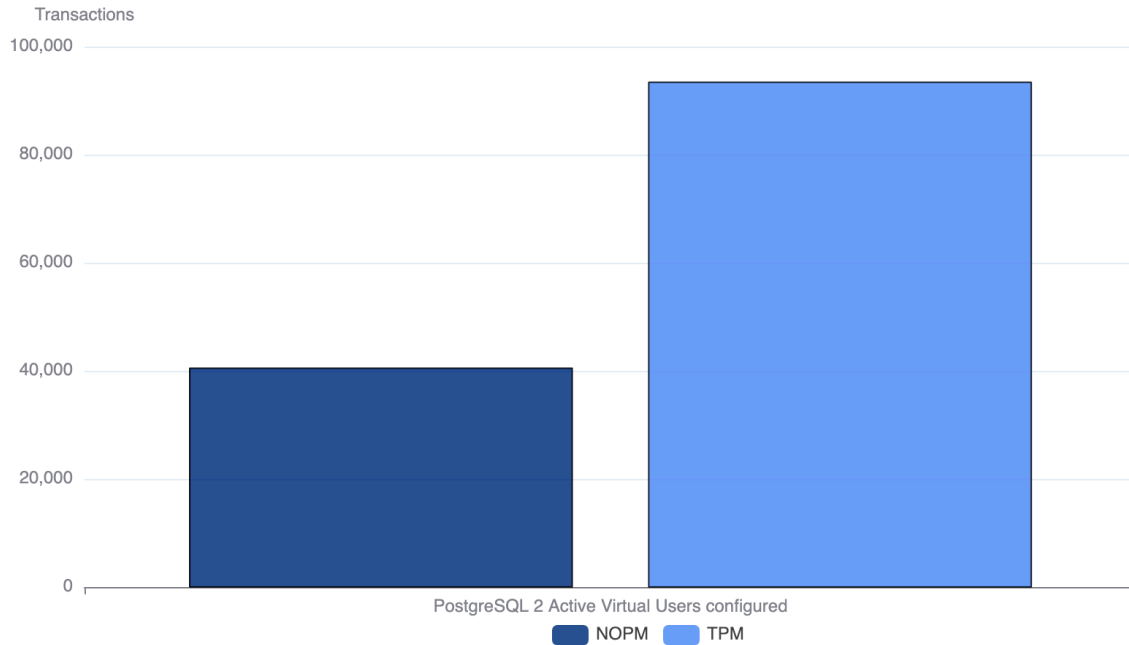
Max_connections = 151

Buffer Pool size = 1G

Log Buffer size = 16MB

HDB Version	Database	Benchmark	NOPM	TPM	Active VU
v5.0	PostgreSQL	TPROC-C	40496	93404	2

Result



Transaction Count



[Job Index](#)

2:

DATES:

18/05/2025

Virtual Users: 5

Where houses: 10

Iterations: 1

NOPM:

TPM:

OS: UBUNTU

Drive: SSD

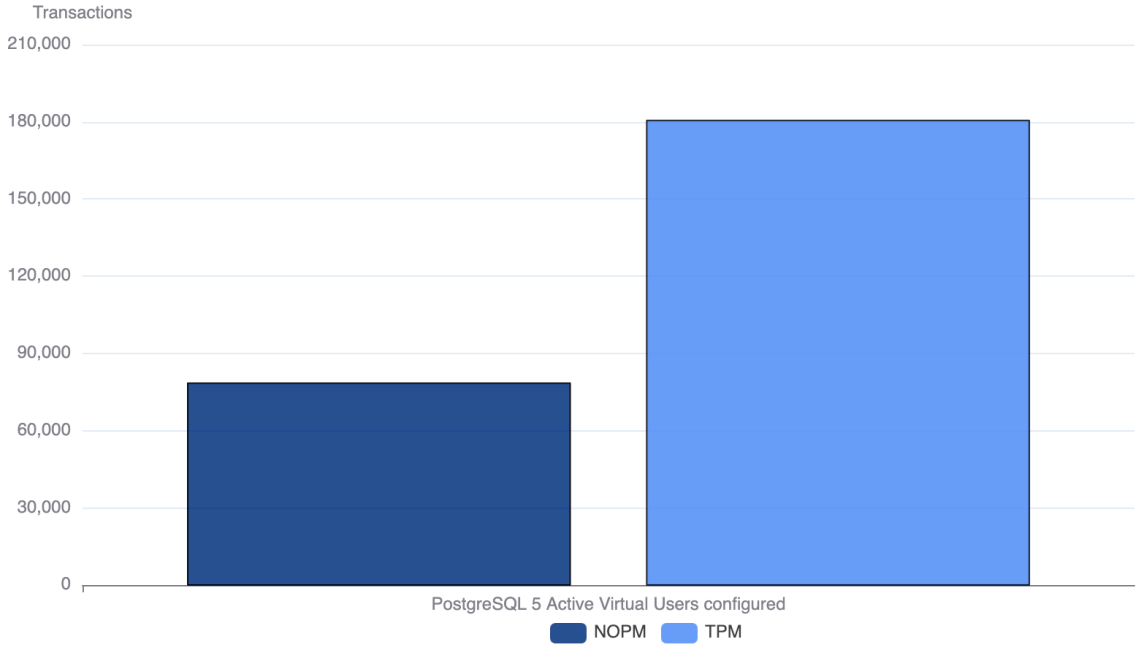
Max_connections = 151

Buffer Pool size = 1G

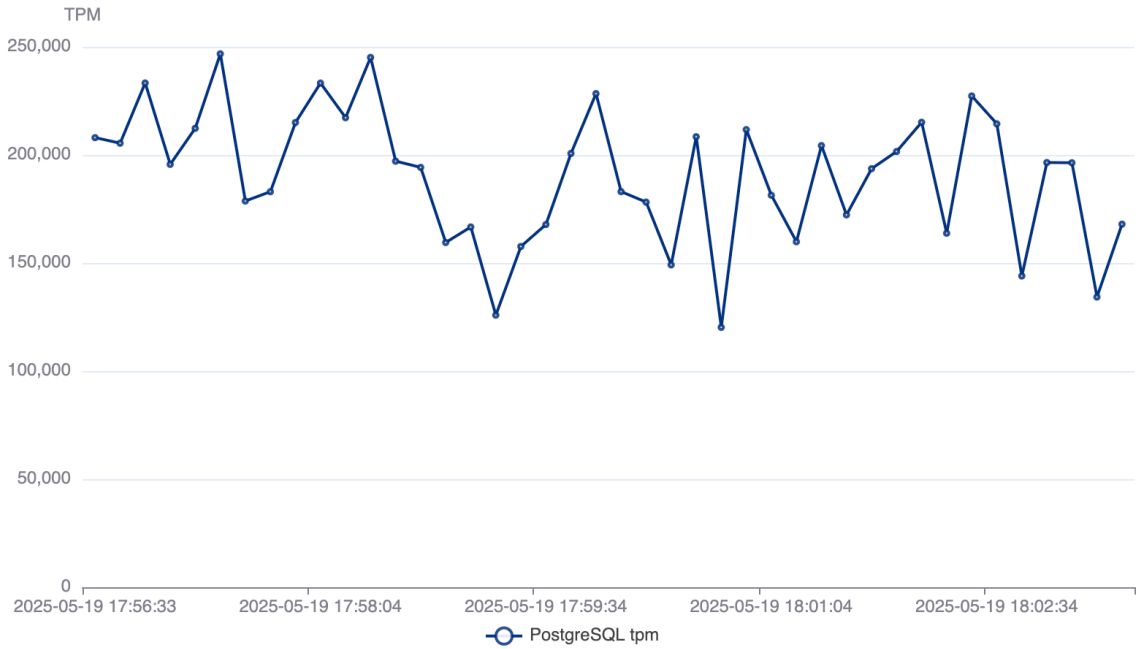
Log Buffer size = 16MB

HDB Version	Database	Benchmark	NOPM	TPM	Active VU
v5.0	PostgreSQL	TPROC-C	78464	180537	5

Result



Transaction Count



[Job Index](#)

3:

DATES:

19/05/2025

Virtual Users: 2

Where houses: 10

Iterations: 1

NOPM:

TPM:

OS: UBUNTU

Drive: SSD

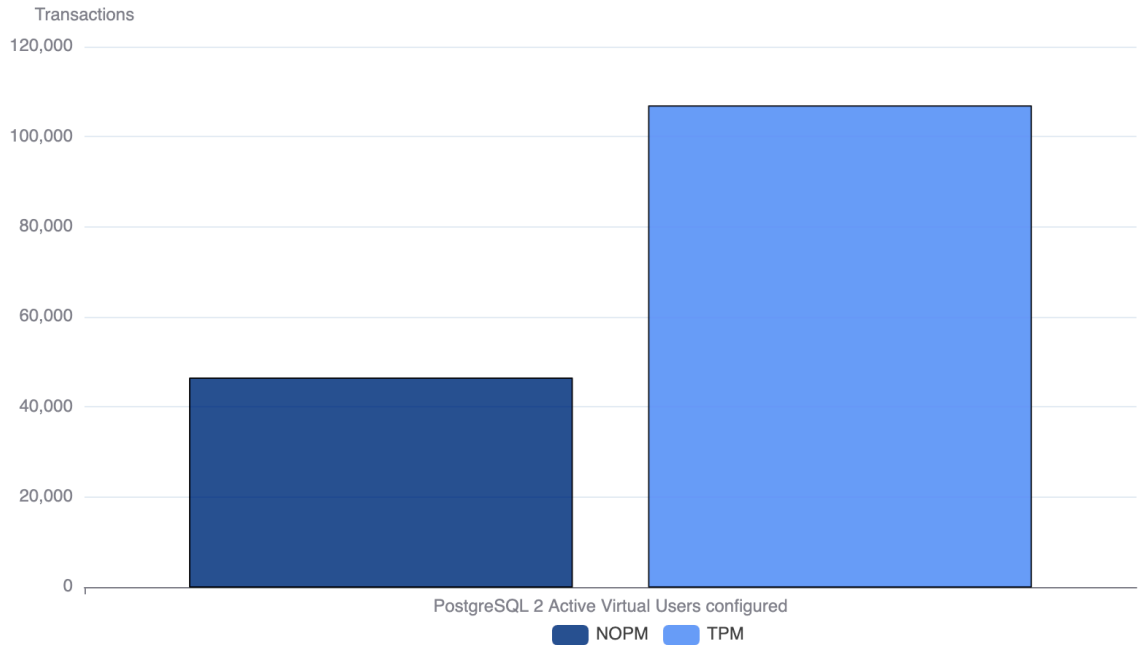
Max_connections = 200

Buffer Pool size = 1G

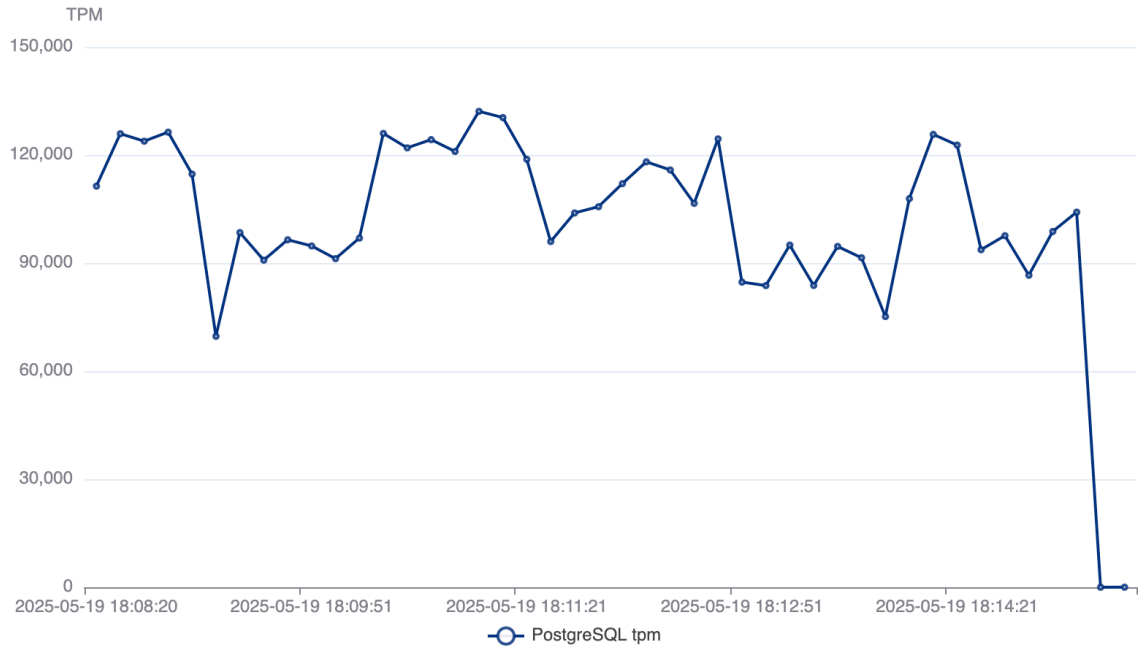
Log Buffer size = 16MB

HDB Version	Database	Benchmark	NOPM	TPM	Active VU
v5.0	PostgreSQL	TPROC-C	46376	106817	2

Result



Transaction Count



[Job Index](#)

4:

DATES:

19/05/2025

Virtual Users: 5

Where houses: 10

Iterations: 1

NOPM:

TPM:

OS: UBUNTU

Drive: SSD

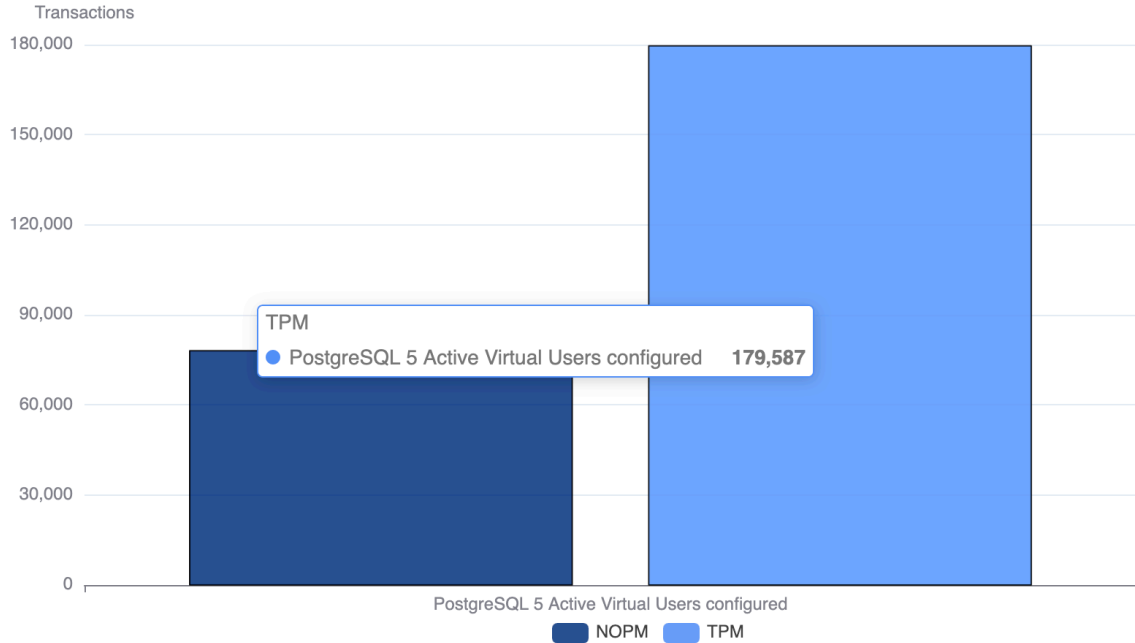
Max_connections = 200

Buffer Pool size = 1G

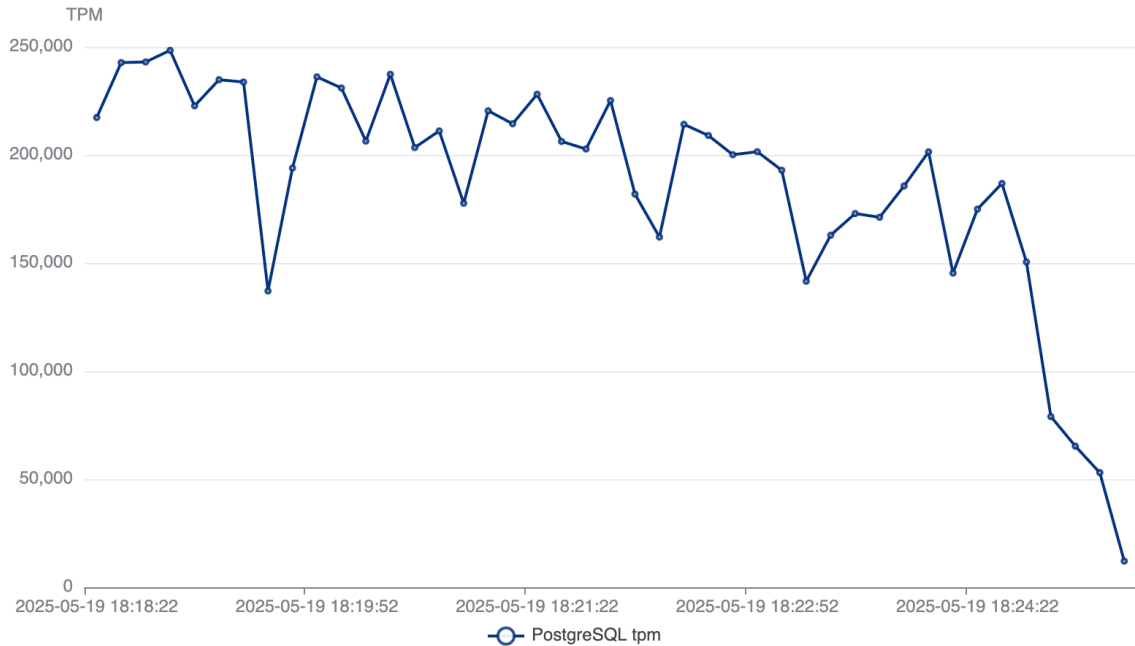
Log Buffer size = 16MB

HDB Version	Database	Benchmark	NOPM	TPM	Active VU
v5.0	PostgreSQL	TPROC-C	78045	179587	5

Result



Transaction Count



[Job Index](#)

Now let's compare with others:

Test Case	max_connections	buffer_pool_size	log_buffer_size	virtual_users
Small	100	256M	8M	5

Test Case	max_connections	buffer_pool_size	log_buffer_size	virtual_users
Medium	200	512M	16M	10
Large	500	1G	32M	20
Max CPU	500	2G	64M	50
I/O Test	200	256M	8M	20
Memory	200	2G	64M	10
Latency	100	128M	8M	5
Stress	1000	4G	128M	100

1:

DATES:

19/05/2025

Virtual Users: 5

Where houses: 5

Iterations: 1

NOPM:

TPM:

OS: UBUNTU

Drive: SSD

Max_connections = 100

Buffer Pool size = 256MB

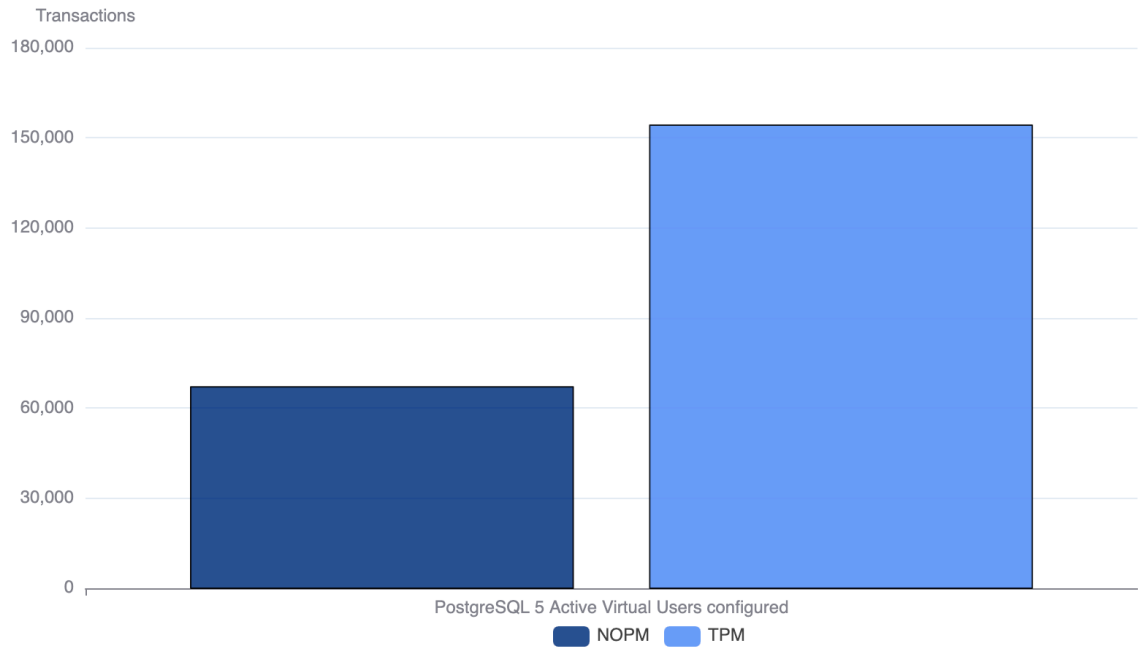
Log Buffer size = 8MB



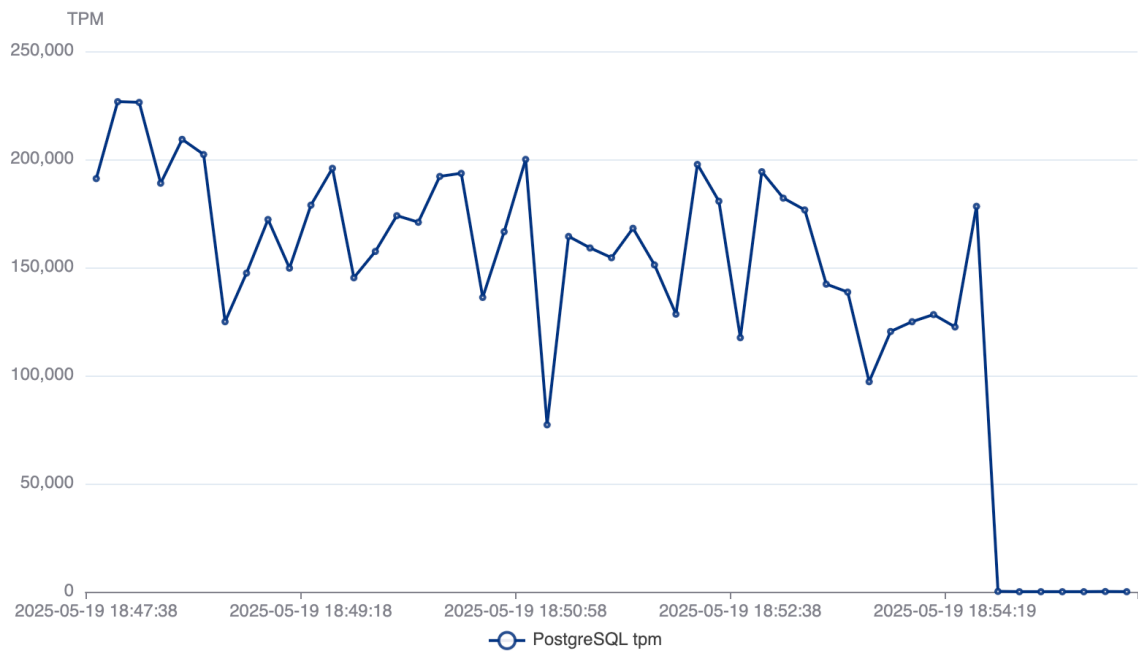
Job 682B6EAE9A103E263430383 TPROC-C Summary 2025-05-19 18:47:26

HDB Version	Database	Benchmark	NOPM	TPM	Active VU
v5.0	PostgreSQL	TPROC-C	67073	154220	5

Result



Transaction Count



[Job Index](#)

2:

DATES:

19/05/2025

Virtual Users: 10

Warehouses: 10

Iterations: 1

NOPM:

TPM:

OS: UBUNTU

Drive: SSD

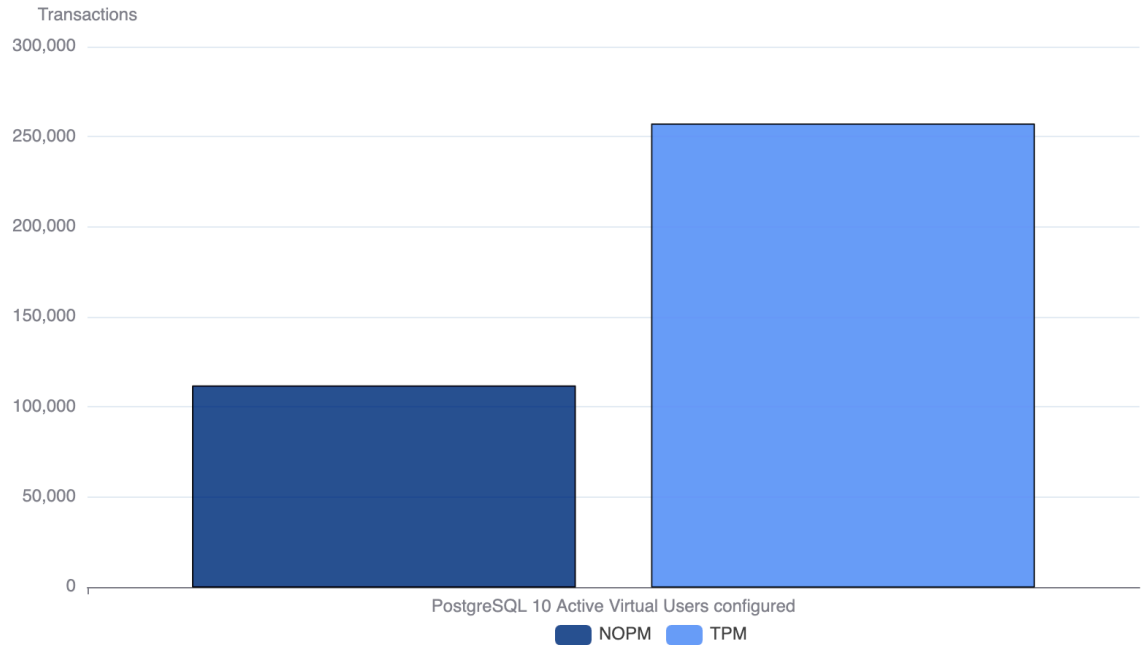
Max_connections = 200

Buffer Pool size = 512MB

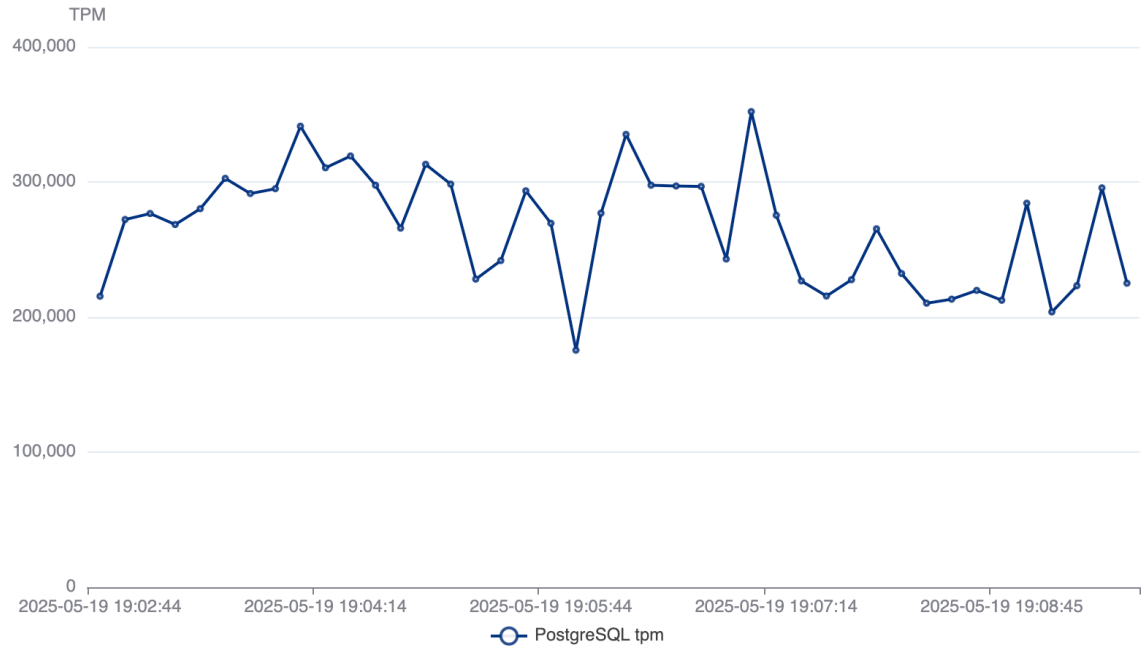
Log Buffer size = 16MB

HDB Version	Database	Benchmark	NOPM	TPM	Active VU
v5.0	PostgreSQL	TPROC-C	111569	256984	10

Result



Transaction Count



[Job Index](#)

3:

DATES:

19/05/2025

Virtual Users: 20

Warehouses: 20

Iterations: 1

NOPM:

TPM:

OS: UBUNTU

Drive: SSD

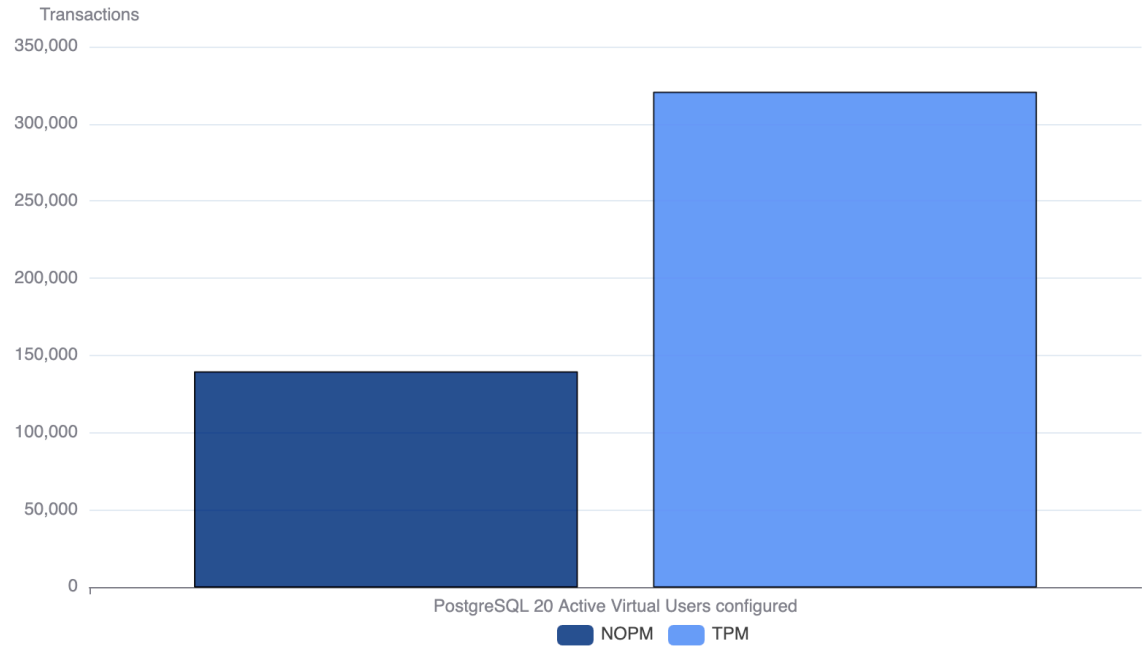
Max_connections = 500

Buffer Pool size = 1G

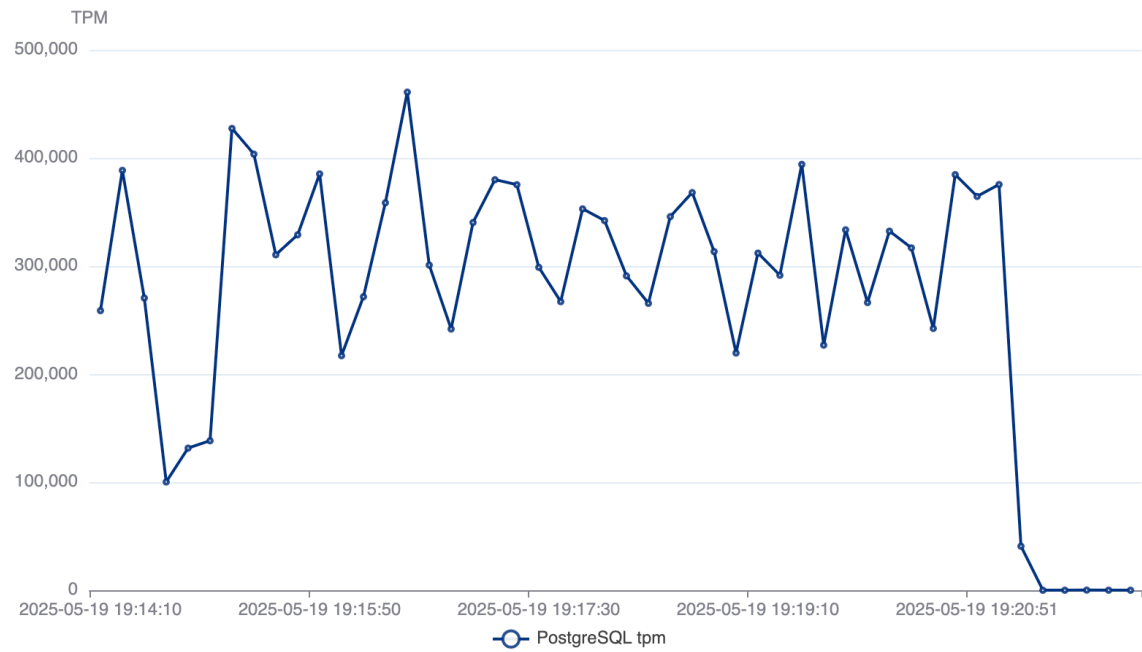
Log Buffer size = 32MB

HDB Version	Database	Benchmark	NOPM	TPM	Active VU
v5.0	PostgreSQL	TPROC-C	139273	320394	20

Result



Transaction Count



[Job Index](#)

4:

DATES:

19/05/2025

Virtual Users: 50

Warehouses: 20

Iterations: 1

NOPM:

TPM:

OS: UBUNTU

Drive: SSD

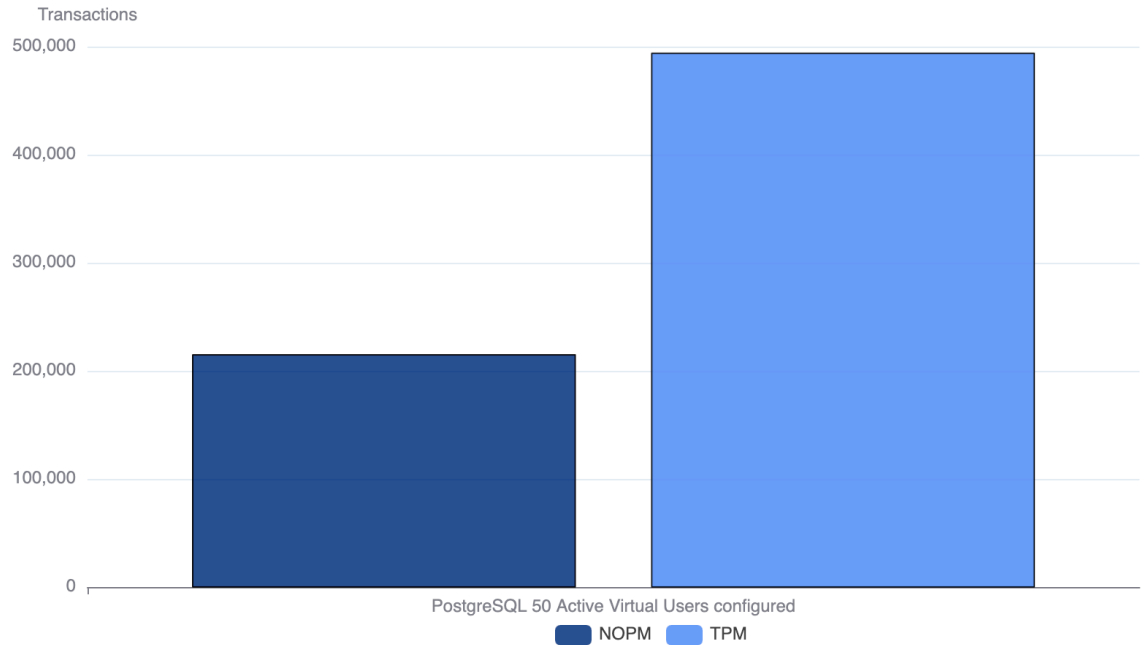
Max_connections = 500

Buffer Pool size = 2G

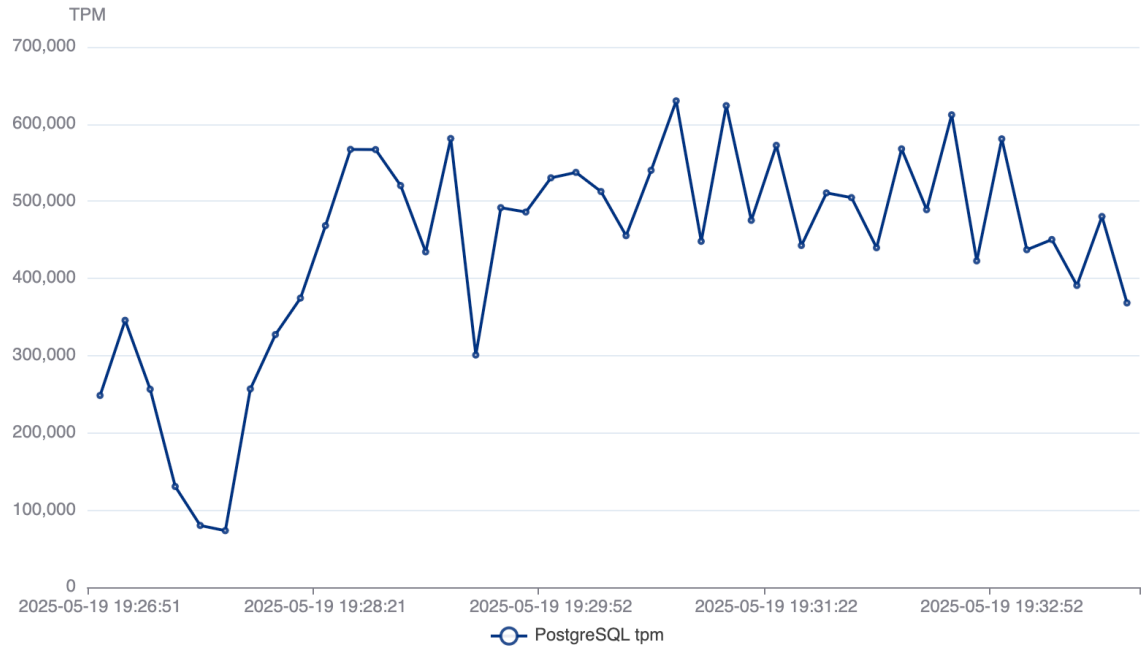
Log Buffer size = 64MB

HDB Version	Database	Benchmark	NOPM	TPM	Active VU
v5.0	PostgreSQL	TPROC-C	214831	493803	50

Result



Transaction Count



[Job Index](#)

5:

DATES:

19/05/2025

Virtual Users: 20

Warehouses: 5

Iterations: 1

NOPM:

TPM:

OS: UBUNTU

Drive: SSD

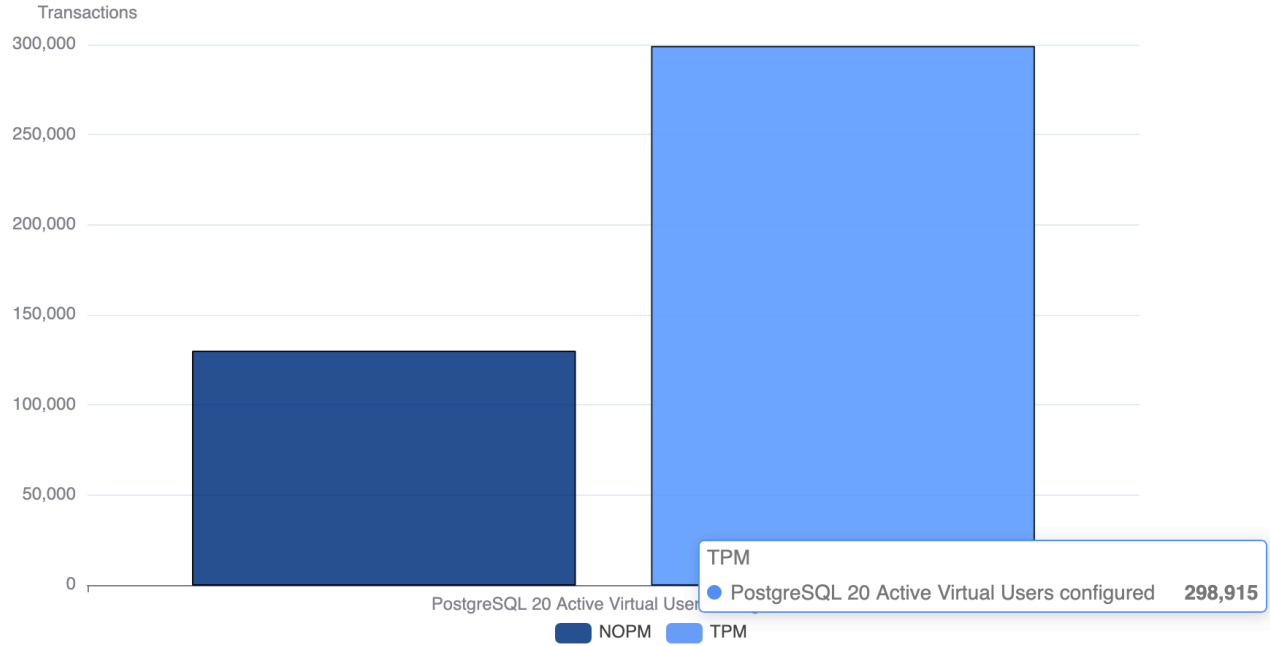
Max_connections = 200

Buffer Pool size = 256MB

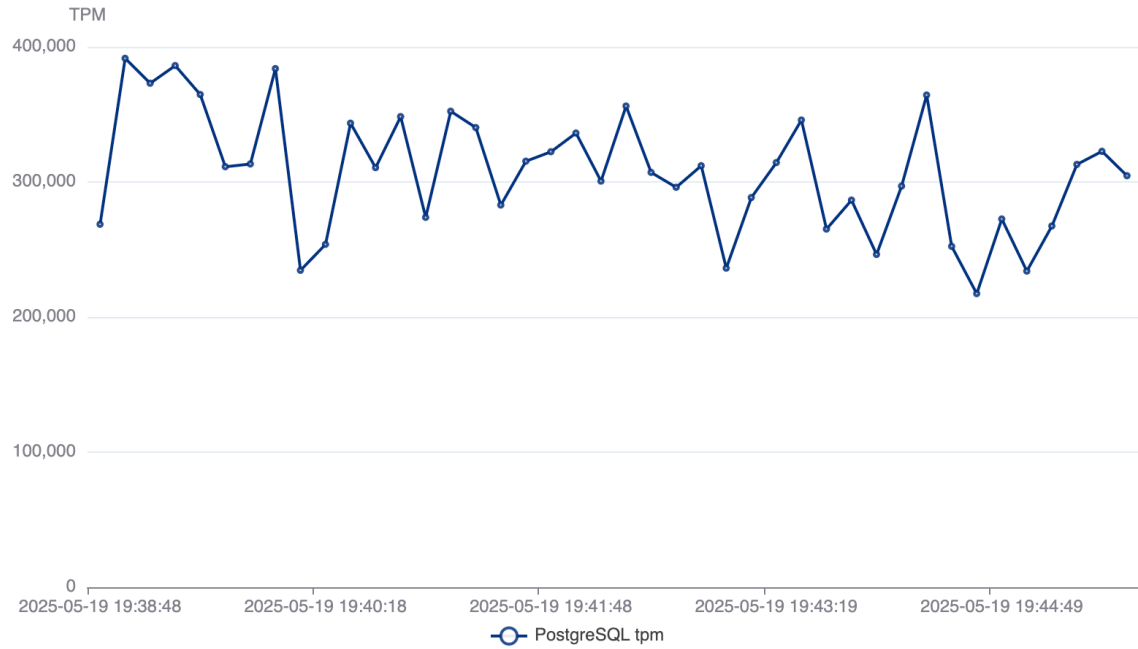
Log Buffer size = 8MB

HDB Version	Database	Benchmark	NOPM	TPM	Active VU
v5.0	PostgreSQL	TPROC-C	129719	298915	20

Result



Transaction Count



[Job Index](#)

6:

DATES:

19/05/2025

Virtual Users: 10

Warehouses: 10

Iterations: 1

NOPM:

TPM:

OS: UBUNTU

Drive: SSD

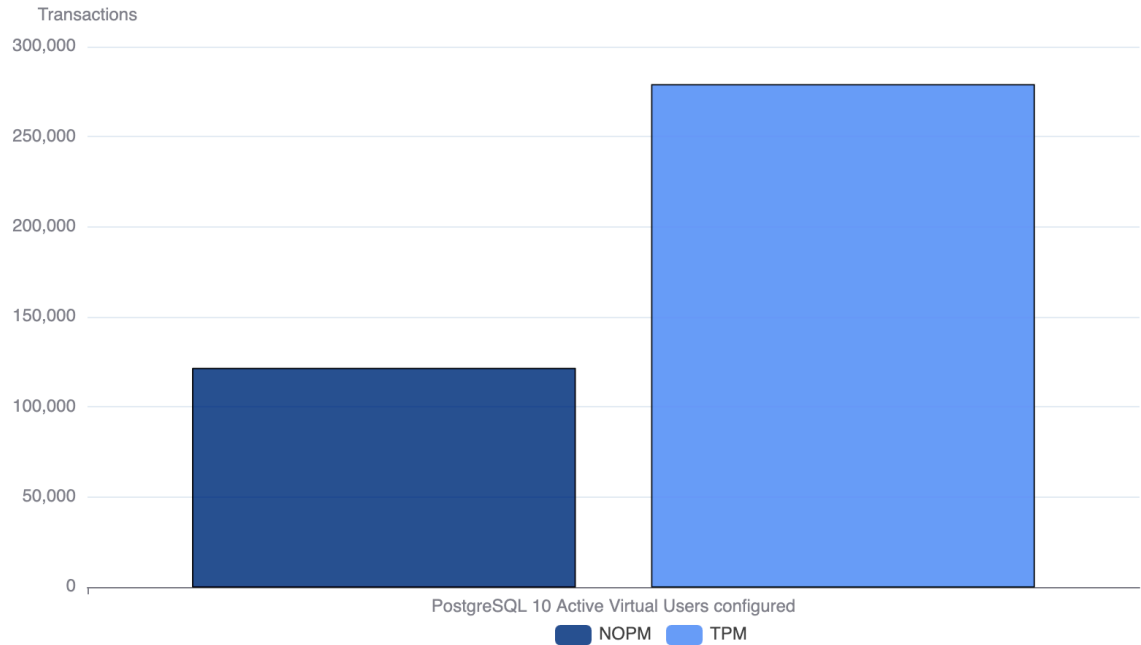
Max_connections = 200

Buffer Pool size = 2G

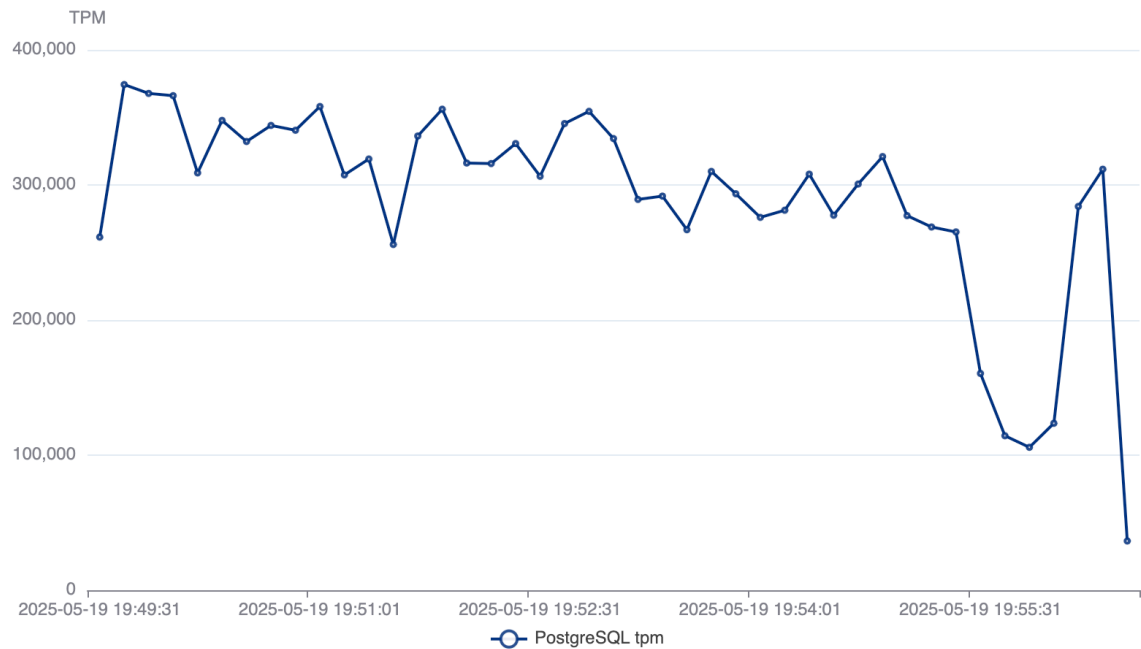
Log Buffer size = 64MB

HDB Version	Database	Benchmark	NOPM	TPM	Active VU
v5.0	PostgreSQL	TPROC-C	121337	278854	10

Result



Transaction Count



[Job Index](#)

7:

DATES:

19/05/2025

Virtual Users: 5

Warehouses: 20

Iterations: 1

NOPM:

TPM:

OS: UBUNTU

Drive: SSD

Max_connections = 100

Buffer Pool size = 128MB

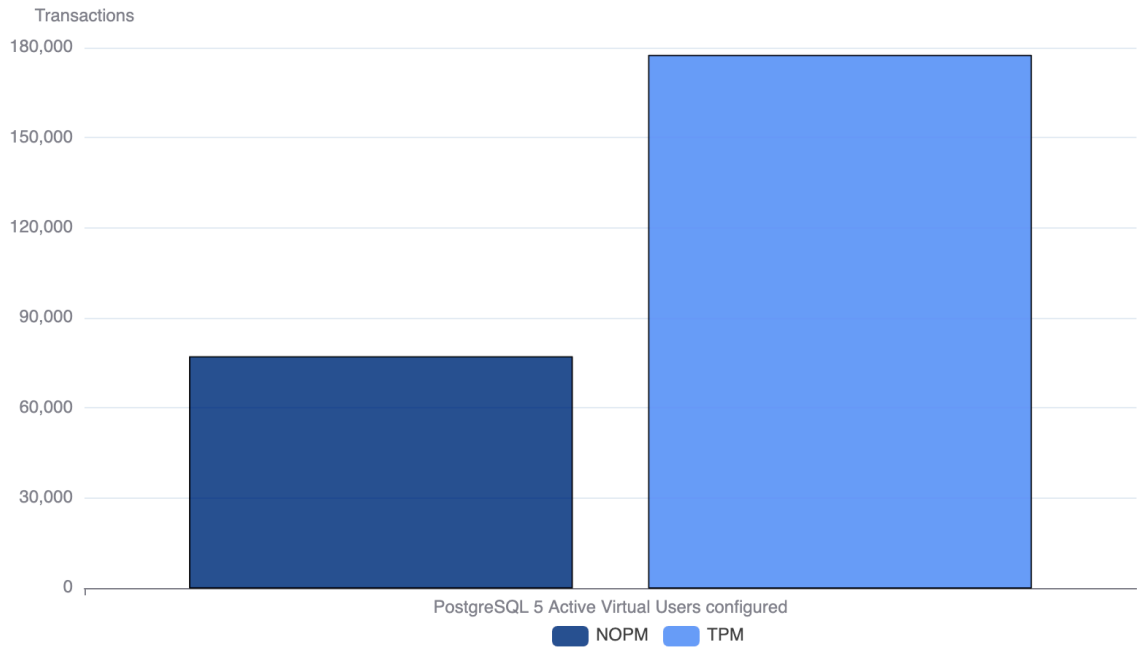
Log Buffer size = 8MB



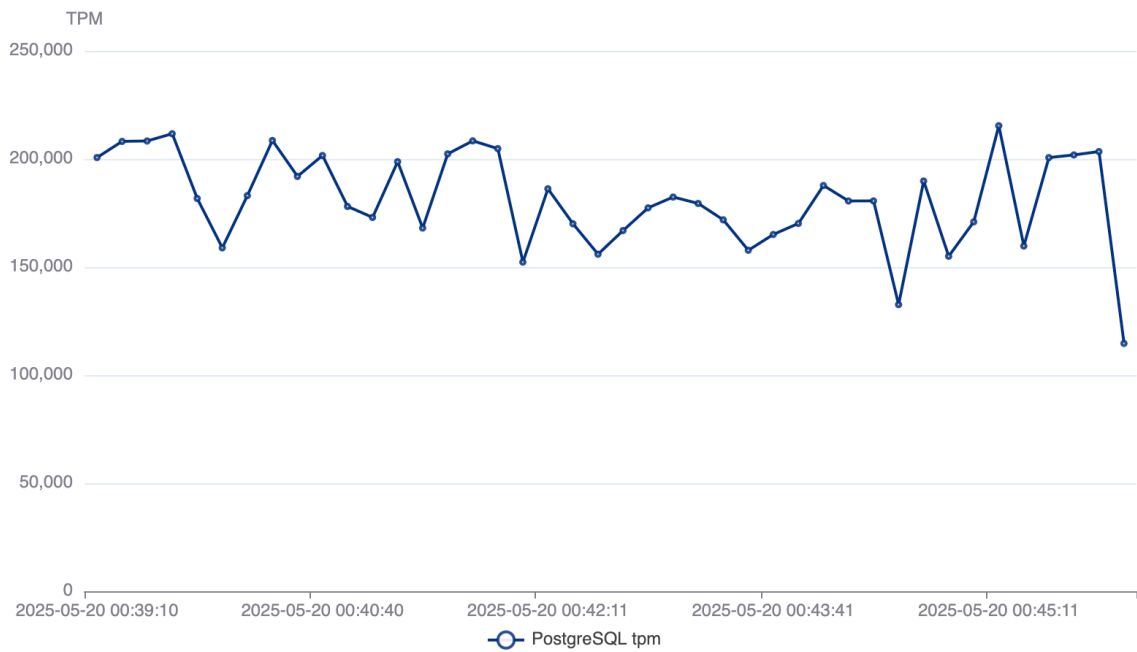
Job 682BC111A2C003E293639353 TPROC-C Summary 2025-05-20 00:38:57

HDB Version	Database	Benchmark	NOPM	TPM	Active VU
v5.0	PostgreSQL	TPROC-C	77007	177388	5

Result



Transaction Count



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8:

DATES:

19/05/2025

Virtual Users: 100

Warehouses: 50

Iterations: 1

NOPM:

TPM:

OS: UBUNTU

Drive: SSD

Max_connections = 1000

Buffer Pool size = 4G

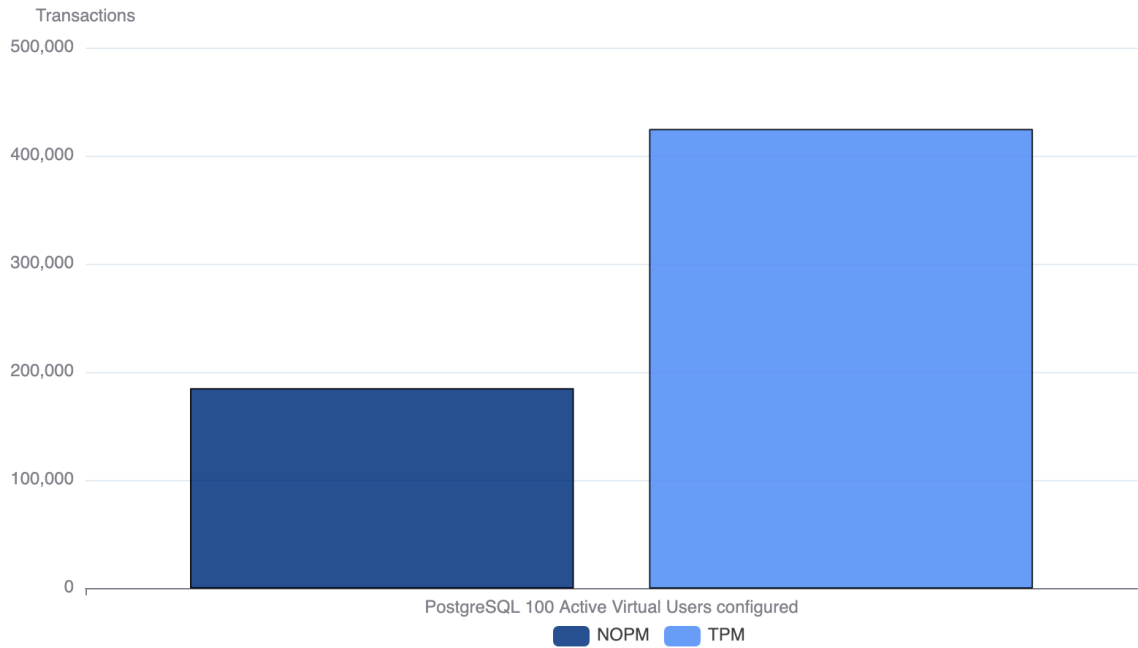
Log Buffer size = 128MB



Job 682BC3D9CD3003E283139333 TPROC-C Summary 2025-05-20 00:50:49

HDB Version	Database	Benchmark	NOPM	TPM	Active VU
v5.0	PostgreSQL	TPROC-C	184345	424191	100

Result



Transaction Count

